



Technical University of Applied Sciences  
Würzburg-Schweinfurt

# Basics of Electronics and Components

## 2. Electric fields

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# Vector calculus

## Lower case greek letters & commonly used capitals

|                          |                        |                   |         |                       |                |
|--------------------------|------------------------|-------------------|---------|-----------------------|----------------|
| $\alpha$                 | alpha                  | $\iota$           | iota    | $\rho$                | rho            |
| $\beta$                  | beta                   | $\kappa$          | kappa   | $\sigma \Sigma$       | sigma          |
| $\gamma \Gamma$          | gamma                  | $\lambda \Lambda$ | lambda  | $\tau$                | tau            |
| $\delta \Delta$          | delta                  | $\mu$             | mu      | $\upsilon \Upsilon$   | upsilon        |
| $\epsilon (\varepsilon)$ | epsilon ('varepsilon') | $\nu$             | nu      | $\phi (\varphi) \Phi$ | phi ('varphi') |
| $\zeta$                  | zeta                   | $\xi \Xi$         | xi      | $\chi$                | chi            |
| $\eta$                   | eta                    | $\omicron$        | omicron | $\psi \Psi$           | psi            |
| $\theta \Theta$          | theta                  | $\pi \Pi$         | pi      | $\omega \Omega$       | omega          |

# Chapter outline

## Learning objectives

Students will be able to:

- Identify the four Maxwell equations in differential and integral forms for systems in equilibrium
- Explain the physical meaning of each equation (Gauss' law, conservative fields, no magnetic monopoles, steady currents)
- Apply Gauss's and Stokes' theorems to convert between integral and differential forms
- Distinguish between electrostatic and magnetostatic equations

## 2.1 Maxwell's equations

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## Learning objectives

Students will be able to:

- Calculate electrostatic quantities ( $Q, \vec{E}, \vec{D}, \varphi, U, \vec{E}, W$ ) and their relationships using vector calculus ( $\vec{E} = -\nabla \varphi, U = \int \vec{E} \cdot d\vec{x}, \vec{D} = Q \vec{E}$ )
- Apply integration techniques to find charge from non-uniform charge densities ( $\lambda, \sigma, \rho$ ) in multiple coordinate systems
- Analyse field configurations using field lines, equipotential surfaces, and path independence
- Derive Maxwell's 1st equation ( $\nabla \cdot \vec{E} = \frac{\rho}{\epsilon_0}$ ) by applying divergence to  $\vec{E} = -\nabla \varphi$  and prove conservative nature ( $\nabla \times \vec{E} = 0$ )

## 2.2 Basic concepts of electric fields

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## Learning objectives

Students will be able to:

- Define  $\vec{D} = \epsilon_0 \epsilon_r \vec{E}$  and explain role of permittivity  $\epsilon$
- Differentiate between polarization (dielectrics: bound charge alignment) and induction (conductors: mobile charge redistribution)
- Explain how relative permittivity  $\epsilon_r$  affects electric field  $\vec{E}$ , capacitance  $C$ , potential  $\varphi$ , and voltage  $U$  while keeping  $D$  constant

## 2.3 Materials

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## Learning objective

Students will be able to:

- Define electric flux ( $\Phi = \oint \vec{D} \cdot d\vec{A} = \int \nabla \cdot \vec{D} dV$ ) and apply Gauss' law ( $\oint \vec{D} \cdot d\vec{A} = Q_{\text{enclosed}}, \nabla \cdot \vec{D} = \rho$ )
- Construct Gaussian surfaces for symmetric problems (spherical, cylindrical, planar) and calculate  $|\vec{E}|$  and  $|\vec{D}|$  without complex integration ( $\oint \vec{D} \cdot d\vec{A} = Q_{\text{enclosed}}$ ). Calculate  $\vec{E}$  and  $\vec{D}$  by exploiting the geometry.
- Analyse electrostatic induction: explain why  $E = 0$  inside conductors and calculate surface charge densities

## 2.4 Electric flux and Gauss' law

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## Learning objective

Students will be able to:

- Energy Storage:
  - Calculate capacitance from geometry using Gauss' law ( $C = \frac{Q}{U} = \frac{Q}{\int \vec{E} \cdot d\vec{x}}$ ) and compute stored energy ( $W = \frac{1}{2} C U^2, W_{\text{vol}} = \frac{1}{2} \epsilon \int |\vec{E}|^2 dV$ )
  - Determine maximum voltage from dielectric strength ( $|\vec{E}| \leq |\vec{E}_{\text{max}}|$ )
- Conductivity:
  - Explain current's microscopic origin ( $\vec{J}_E = \mu \vec{D}, \vec{J} = n(Q) \vec{v}_d = \kappa \vec{E}$ ) and derive Ohm's law ( $R = \frac{U}{I} = \rho \frac{L}{A}$ ) from field theory
  - Calculate conductor properties ( $\mu, \epsilon_0, \kappa, \rho$ ) from material parameters

## 2.5 Field applications: energy storage and conductivity

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# 2. Electric fields

## Learning objectives

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- Distinguish between electrostatic and magnetostatic equations

# 2.1 Maxwell's equations

# 2.1 Maxwell's equations

## Differential form

|      | Formula                                                                       | Name and physical meaning                                                                                                                                                                       |
|------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| I.   | $\vec{\nabla} \cdot \vec{D} = \rho$                                           | <b>Gauss' law:</b> “The net electric flux $\vec{\nabla} \cdot \vec{D}$ is proportional to the charge density $\rho$ at that same point.”                                                        |
| II.  | $\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$          | <b>Faraday's law of induction:</b> “a changing magnetic field $\frac{\partial \vec{B}}{\partial t}$ creates a circulating electric field $\vec{\nabla} \times \vec{E}$ .”                       |
| III. | $\vec{\nabla} \cdot \vec{B} = 0$                                              | <b>Gauss' law of magnetics:</b> “the magnetic field has zero divergence, meaning magnetic field lines never begin or end at a point (no isolated magnetic poles).”                              |
| IV.  | $\vec{\nabla} \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t}$ | <b>Ampère–Maxwell equation:</b> “current density $\vec{J}$ and a time-varying flux density $\frac{\partial \vec{D}}{\partial t}$ and produces a magnetic field $\vec{\nabla} \times \vec{H}$ .” |

# 3.1 Maxwell's equations

## Systems in equilibrium

### Formula

### Name and physical meaning

#### Electrostatics

I.  $\vec{\nabla} \cdot \vec{D} = \rho$

**Gauss' law:** "The net electric flux  $\vec{\nabla} \cdot \vec{D}$  is proportional to the charge density  $\rho$  at that same point."

II.  $\vec{\nabla} \times \vec{E} = 0$

**Limit of Faraday's law:** "In equilibrium there is no induced electric field." (conservative field)

#### Magnetostatics

III.  $\vec{\nabla} \cdot \vec{B} = 0$

**Gauss' law of magnetics:** "There are no free magnetic poles."

IV.  $\vec{\nabla} \times \vec{H} = \vec{j}$

**Ampère's law:** "Electric currents create steady magnetic field loops around them." (right hand rule)

# 2.1 Maxwell's equations

## Comparing integral and differential form in equilibrium

### Integral form

I.  $\oiint_{\partial\Omega} \vec{D} \cdot d\vec{A} = \iiint_{\Omega} \vec{\nabla} \cdot \vec{D} dV = Q_{\text{encl.}}$

II.  $\oint_{\partial\Sigma} \vec{E} \cdot d\vec{l} = \iint_{\Sigma} \vec{\nabla} \times \vec{E} \cdot d\vec{A} = 0$

III.  $\oiint_{\partial\Omega} \vec{B} \cdot d\vec{A} = \iiint_{\Omega} \vec{\nabla} \cdot \vec{B} dV = 0$

IV.  $\oint_{\partial\Sigma} \vec{H} \cdot d\vec{l} = \iint_{\Sigma} \vec{\nabla} \times \vec{H} \cdot d\vec{A} = I$

### Differential form

$$\vec{\nabla} \cdot \vec{D} = \rho$$

$$\vec{\nabla} \times \vec{E} = 0$$

$$\vec{\nabla} \cdot \vec{B} = 0$$

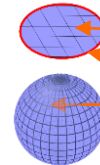
$$\vec{\nabla} \times \vec{H} = \vec{j}$$



Using Gauss's and Stokes's theorems, we can convert between integral and differential forms. See chapter 1 for the mathematical details.

Gauss' law

Stokes' law



$\Sigma$ : surface  
 $\partial\Sigma$ : boundary curve  
 $\Omega$ : volume  
 $\partial\Omega$ : boundary surface

## Summary: Maxwell's Equations in Equilibrium:

- Electrostatics:
  - └ Gauss' law:  $\vec{\nabla} \cdot \vec{D} = \rho \rightarrow$  "Charges create flux"
  - └ Conservative:  $\vec{\nabla} \times \vec{E} = 0 \rightarrow$  „Do not rotate and path-independent"
- Magnetostatics:
  - └ No monopoles:  $\vec{\nabla} \cdot \vec{B} = 0 \rightarrow$  "Closed loops"
  - └ Ampère:  $\vec{\nabla} \times \vec{H} = \vec{j} \rightarrow$  "Currents create fields"
- Conversion: Gauss' & Stokes' theorems link integral  $\leftrightarrow$  differential forms via del operator

## 2.1 Maxwell's equations

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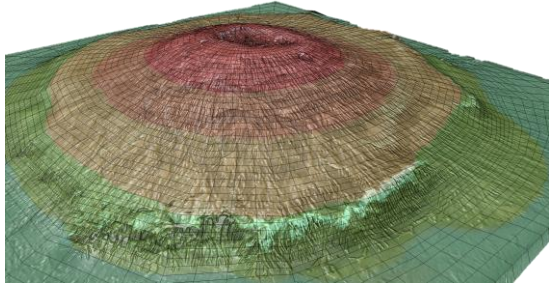
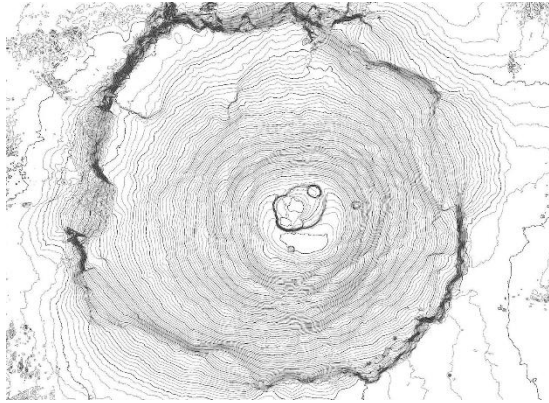
## 2.2 Basic concepts of electric fields

## 2.2 Basic concepts of electric field

### Analogy between a landscape and electric potential

Quick check

Think of a marble rolling on a landscape. Which features of the topographic map make it roll faster? In the electric-potential analogy, what would these features correspond to?

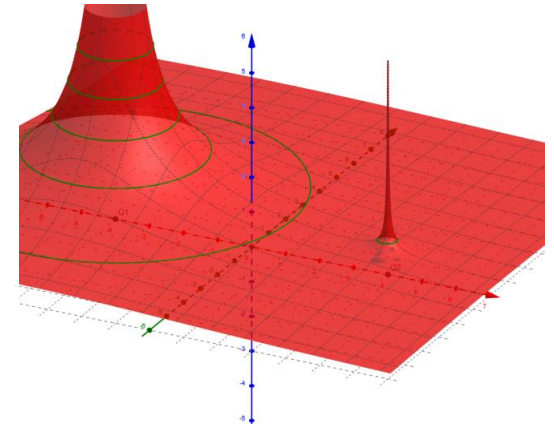
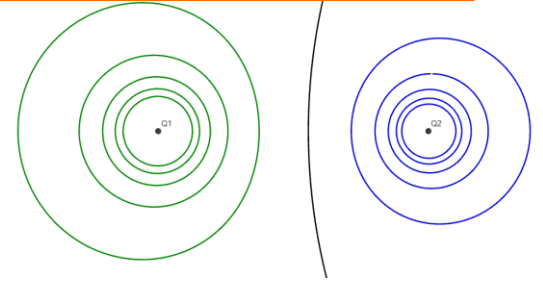


Topographic map (**top**) and 3D model (**bottom**) Olympus Mons – Mars.

Landscape  
(scalar field)



Potential  
'landscape'  
(scalar field)



Equipotential lines of two charges (**top**) and 3D model (**bottom**).

# 2.2 Basic concepts of electric fields

## Electric charge

- **Electric charge**  $Q$  is the physical property of matter. Its electromagnetic interaction is the **cause for magnetic and electric fields**.

- The **unit** of charge is the **Coulomb**.

$$[Q] = C = A \cdot s$$

- **Properties** of electric charges

- ⌋ Can be **positive** (i. e. protons) or **negative** (i. e. electrons)
- ⌋ **Like** charges **repel**; **unlike** charges **attract** each other
- ⌋ Charge is **quantified**. The base unit is the **elementary charge**:

$$e = 1.602 \cdot 10^{-19} \text{ C}$$

( $6.241 \cdot 10^{18}$  electrons are equivalent to a charge of  $-1 \text{ C}$ )

- ⌋ Cause for **electric potential** and **electric fields**
- ⌋ **Moving** charges generate **magnetic fields**
- ⌋ **Is conserved**: it can neither be generated nor destroyed



We use  $Q$  when talking about **source / total charge** and  $q$  when talking about a **test charge**



*Eponymy for electricity: The ancient greek word for amber, ἤλεκτρον (ēlektron).*

## 2.2 Basic concepts of electric fields

### Quantities derived from the electric charge

#### — Electric force

- 7 Definition: Coulomb force between two charges.
- 7 Intuition: Like charges repel, unlike charges attract each other.

#### — Electric field

- 7 Definition: Force per unit test charge at a point in space.
- 7 Intuition: Force field around a charge, showing how it attracts or repels other charges.

#### — Electric flux

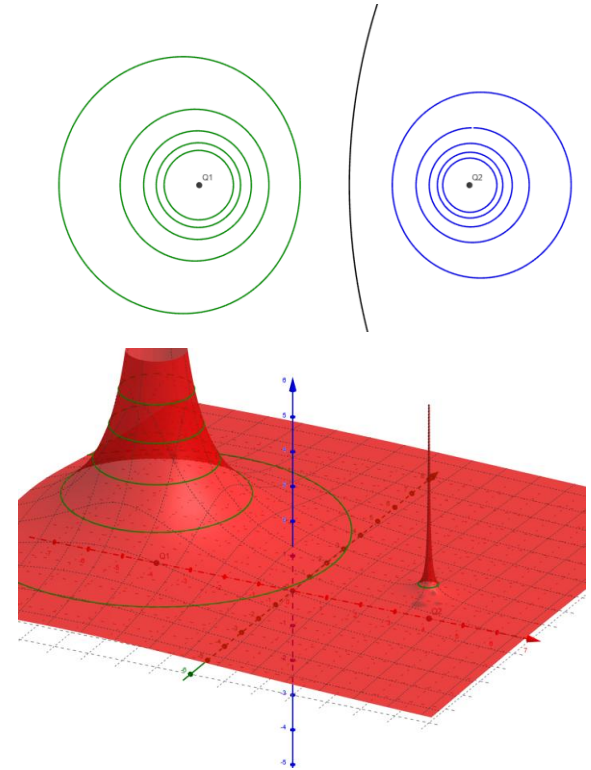
- 7 Definition: Surface integral of the electric field through an area.
- 7 Intuition: Number of electric field lines passing through a surface

#### — Electric potential

- 7 Definition: Energy per unit charge at a point in the field.
- 7 Intuition: Electric energy level of a location in the field.

#### — Voltage

- 7 Definition: Potential difference between two points
- 7 Intuition: Work per charge required or released when moving charge between two points.



*Equipotential lines of two charges (top) and 3D model (bottom).*

# 2.2 Basic concepts of electric fields

## Charge density

- **Charge density** is charge per unit of length, area or volume.
- Use it to calculate the charge  $Q$  of a body, e. g. a sphere.
- $\mathbb{R}^1$ : constant **line** charge density  $\lambda$  of a length  $R$ :

$$dQ = \lambda dr$$

$$Q = \lambda \int_{r=0}^R dr = \lambda R$$

- $\mathbb{R}^2$ : constant **surface** charge density  $\sigma$  of a circle:

$$dQ = \sigma dA = \sigma r d\phi dr$$

$$Q = \sigma \int_{r=0}^R \int_{\phi=0}^{2\pi} r d\phi dr = \sigma \pi R^2$$

- $\mathbb{R}^3$ : constant **space** charge density  $\rho$  of a sphere:

$$dQ = \rho dV = \rho r d\theta r \sin \theta d\phi dr$$

$$Q = \rho \int_{r=0}^R \int_{\phi=0}^{2\pi} \int_{\theta=0}^{\pi} r^2 \sin \theta d\theta d\phi dr = \rho \frac{4}{3} \pi R^3$$

- *Note: In case of a non-constant charge density, it must be integrated, as well.*

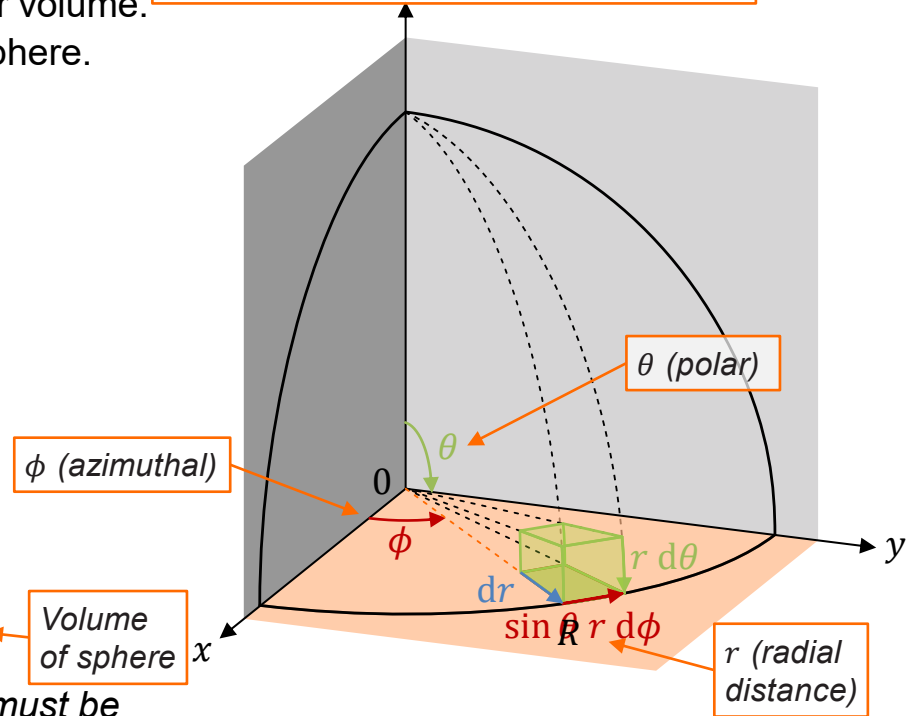


Units of charge densities:

$\mathbb{R}^1$ :  $[\lambda] = \text{C m}^{-1}$  (lambda: 'length')

$\mathbb{R}^2$ :  $[\sigma] = \text{C m}^{-2}$  (sigma: 'surface')

$\mathbb{R}^3$ :  $[\rho] = \text{C m}^{-3}$  (rho: 'room')

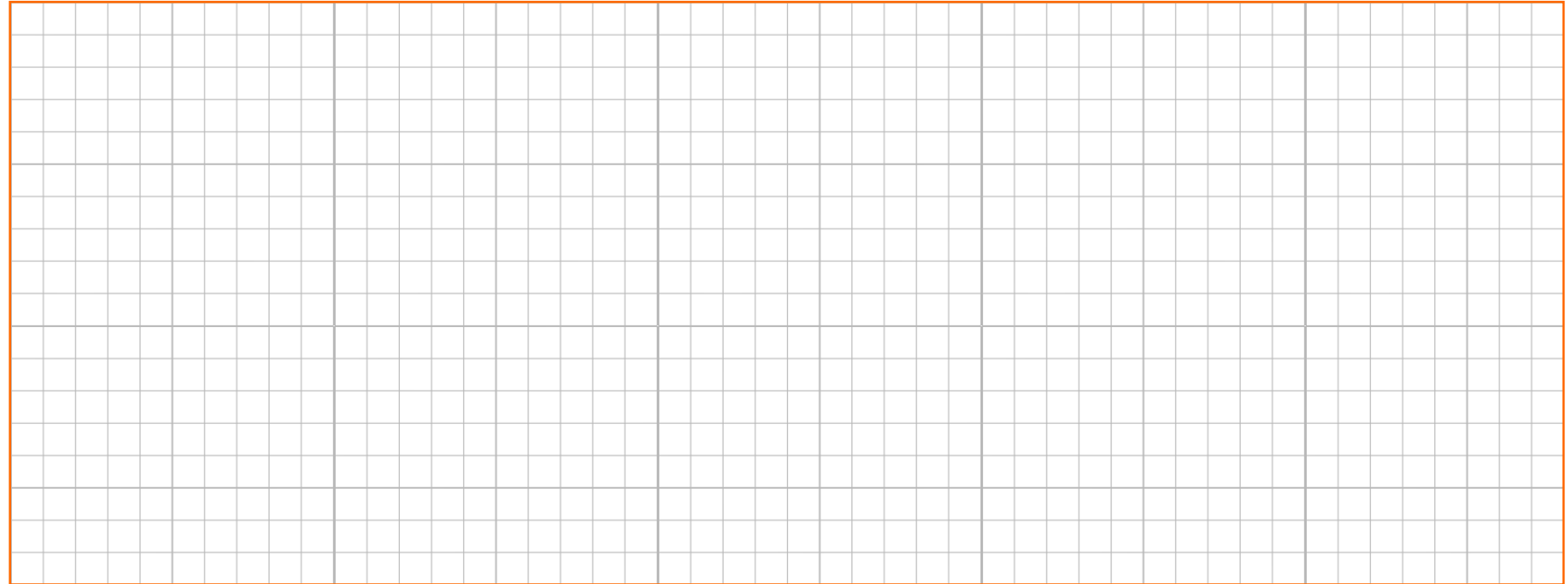


Cross section of a sphere in polar coordinates.

## 2.2 Basic concepts of electric fields

### Exercise: variable charge density

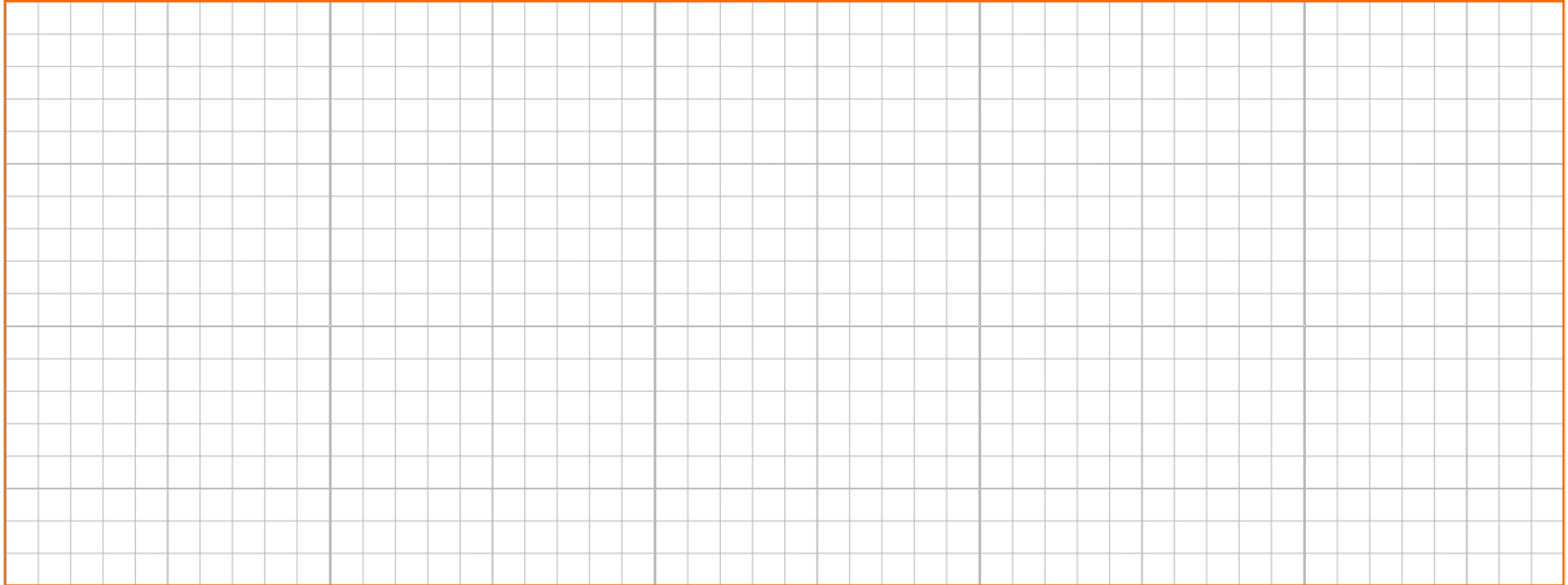
Calculate the charge  $Q$  of a sphere with radius  $R = 1$  m with linearly decaying charge density  $\rho = \frac{\rho_0}{r}$ , where  $\rho_0 = 160 \text{ mC m}^{-2}$  is an arbitrary constant. Round the result to two significant digit.



## 2.2 Basic concepts of electric fields

### Homework: Charge density

Derive the formulas for the line, surface and space charge, assuming the charge densities are constants.



## 2.2 Basic concepts of electric fields

### Coulomb force

- The force  $\vec{F}_{12}$  on a charge  $Q_1$  attracted by another charge  $Q_2$

$$\vec{F}_{12} = \frac{1}{4 \pi \varepsilon} \frac{Q_1 Q_2}{r_{12}^2} \vec{e}_{12} = \frac{1}{4 \pi \varepsilon} \frac{Q_1 Q_2}{r_{12}^3} \vec{r}_{12} = -\vec{F}_{21}$$

where

- ⌈ The unit of the force is in Newton  $[F] = \text{N} = \frac{\text{kg} \cdot \text{m}}{\text{s}^2}$
- ⌈ Unit vector  $\vec{e}_{12} = \frac{\vec{r}_{12}}{r_{12}}$  from point  $Q_1$  to  $Q_2$  and  $r_{12} = \|\vec{r}_{12}\|$
- ⌈ The permittivity  $\varepsilon = \varepsilon_0 \varepsilon_r$  and the vacuum permittivity

$$\varepsilon_0 = \frac{1}{4 \pi 10^{-7} c^2} = 8.854\,187\,817 \cdot 10^{-12} \frac{\text{As}}{\text{Vm}}$$

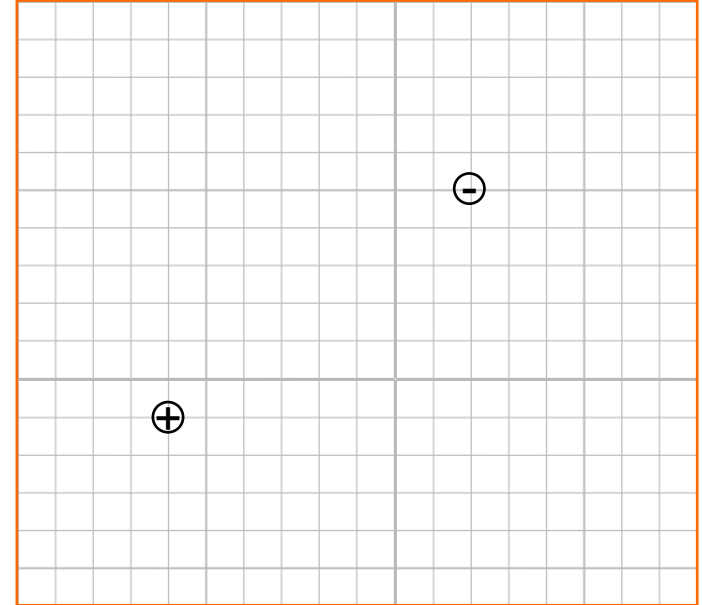
where  $c = 299\,792\,458 \text{ m/s}$  is the speed of light

- ⌈ Relative permittivity  $\varepsilon_r \geq 1$  (no unit) depending on the degree of polarization of the medium. For vacuum  $\varepsilon_r = 1$ .
- The magnitude of the Coulomb force is

$$\|\vec{F}_{12}\| = \frac{1}{4 \pi \varepsilon} \frac{|Q_1| |Q_2|}{r_{12}^2}$$

*Polar to rectangular via Pythagorean in 2D*


$$\begin{aligned} r_{12} &= ((x_2 - x_1)^2 + (y_2 - y_1)^2)^{1/2} \\ r_{12}^2 &= (x_2 - x_1)^2 + (y_2 - y_1)^2 \\ r_{12}^3 &= ((x_2 - x_1)^2 + (y_2 - y_1)^2)^{3/2} \end{aligned}$$



*An electric charge  $Q_1$  attracts another charge  $Q_2$  and vice versa.*

## 2.2 Basic concepts of electric fields

### Electric field

- To describe the **influence of a charge**  $Q$  without specifying a particular test charge  $q_{\text{test}}$  we define the **electric field**  $\vec{E}$

$$\vec{E}(\vec{r}) = \frac{\vec{F}(\vec{r})}{q_{\text{test}}} = \frac{1}{4\pi\epsilon} \frac{Q}{r^2} \vec{e}_r \quad [E] = \frac{\text{N}}{\text{C}} = \frac{\text{V}}{\text{m}}$$

- Its magnitude is

$$\|\vec{E}(\vec{r})\| = \frac{1}{4\pi\epsilon} \frac{|Q|}{r^2}$$

- For many charges, fields **superpose**:

$$\vec{E}(\vec{r}) = \frac{1}{4\pi\epsilon} \sum_i Q_i \frac{\vec{r}_i}{r_i^3} = \frac{1}{4\pi\epsilon} \int_V \rho \frac{\vec{r}}{r^3} dV$$



By dividing the force by  $q_{\text{test}}$  the field becomes a **property of the source charge and of space** (the medium's permittivity), not of any particular charge pair.

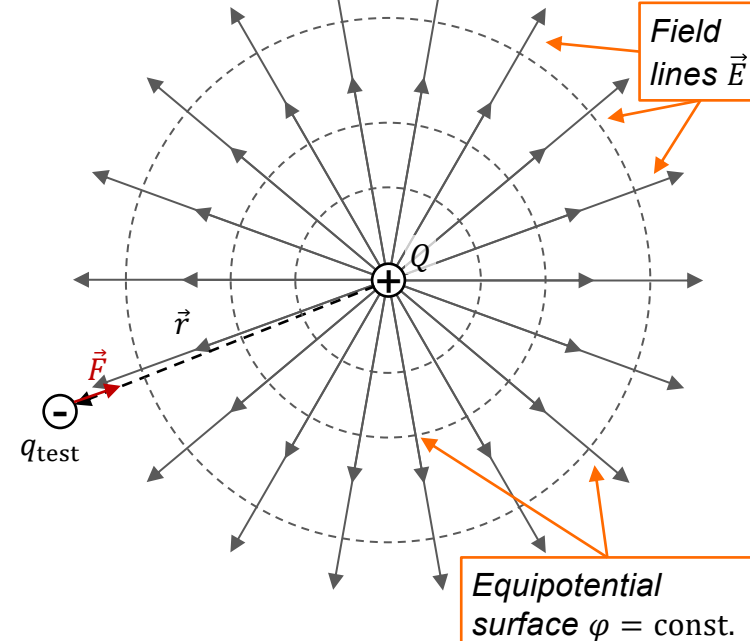


We'll use the same 'trick' later, dividing the electric field by the permittivity  $\epsilon$  so that the flux density becomes a **property of the charge**, not of the surrounding space.

?

Quick  
check

How do (a) the force on  $q_{\text{test}}$  and (b) the electric field change when you ...  
1: ... double the test charge  $q_{\text{test}}$ .  
2: ... move the original test charge into a medium with twice the permittivity.



Electric field and equipotential surfaces around a point charge  $Q$ .

## 2.2 Basic concepts of electric fields

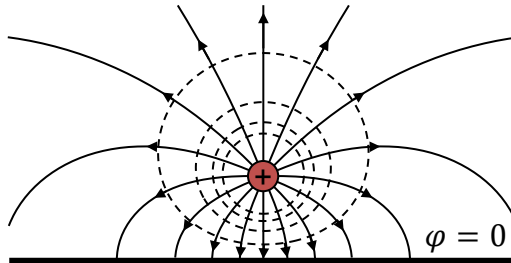
### Visualizing fields and potentials

**Equipotential surfaces:** Surfaces of constant electric field strength represented by closed curves.

#### Field lines

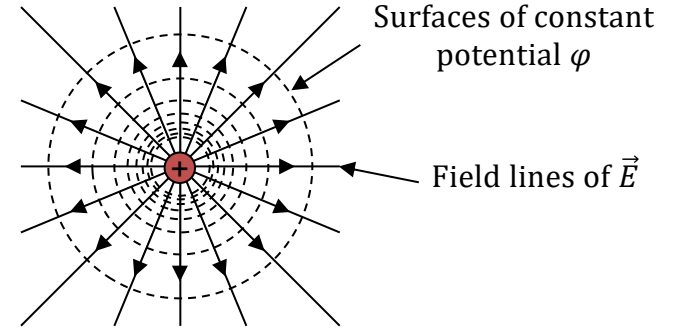
- Point from positive to negative potentials
- Are tangent to  $\vec{E}$  field, do not cross
- Perpendicular to equipotential surfaces
- 'Line density' is equivalent strength of field

*Warning: no overlay of field-images!*

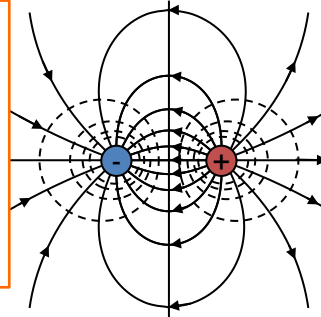


*Field lines and equipotential lines of a point charge and a metallic plate*

 Field lines from different sources cannot simply be superimposed visually because line density represents field strength. Instead, calculate the vector sum of fields and then draw the resulting field lines.



*Field lines and equipotential lines (representing a 3D sphere) of a point charge.*

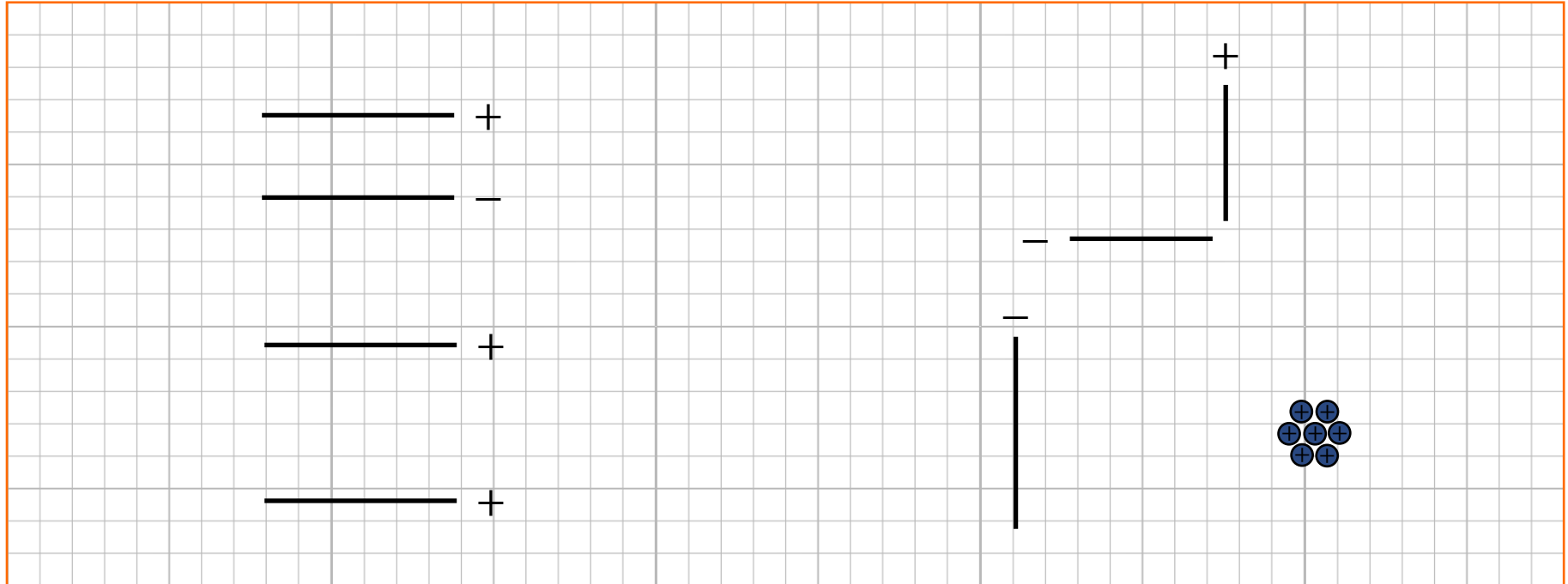


*Field lines and equipotential lines of two point charges of opposite charge.*

## 2.2 Basic concepts of electric fields

### Exercise: Field lines

Draw the electrical field lines and equipotential lines for the following setups:



## 2.2 Basic concepts of electric fields

Example: Electric field of a wire. 1/2

Euclidian norm in 2D

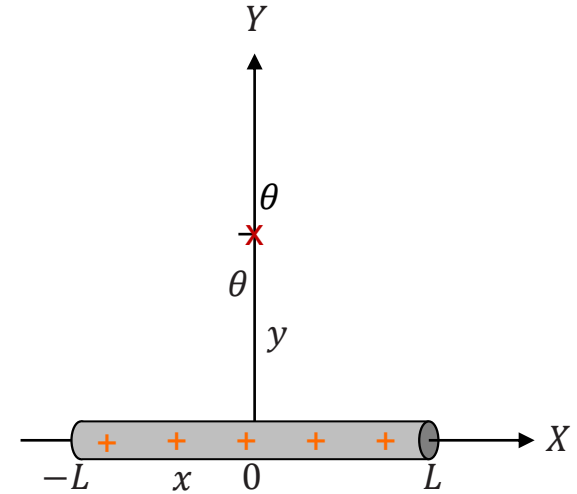


$$r = (x^2 + y^2)^{1/2}$$

$$r^2 = x^2 + y^2$$

$$r^3 = (x^2 + y^2)^{3/2}$$

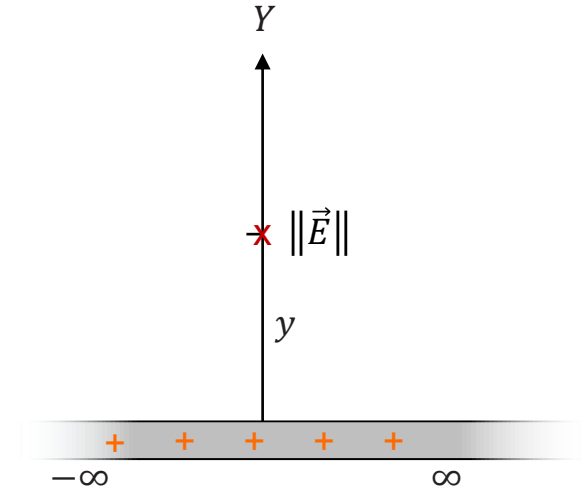
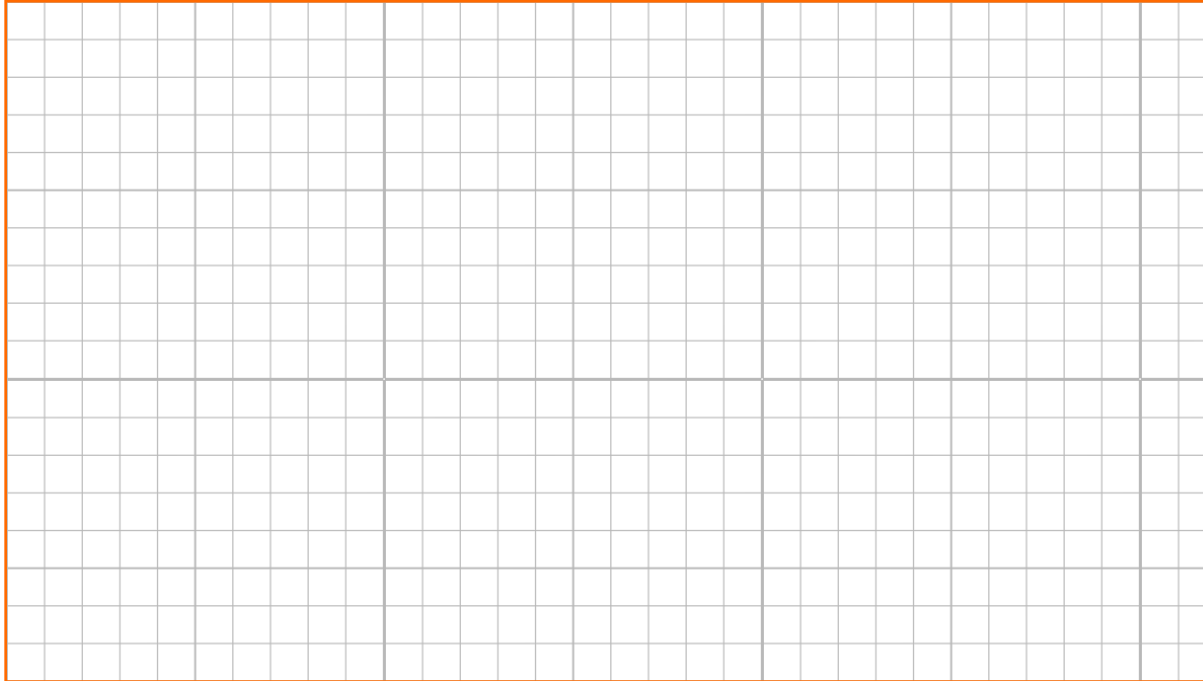
Calculate the magnitude of the electric field  $\|\vec{E}\|$  at distance  $y$  from a wire of length  $-L \leq x \leq L$  and charge density  $\lambda$ . Calculate the electric field  $\vec{E}$ .



## 2.2 Basic concepts of electric fields

Example: Electric field of a wire. 2/2

Now expand the length of the line towards  $\pm\infty$ . Calculate the magnitude of the electric field.



## 2.2 Basic concepts of electric fields

### Electric work and potential

- **Work**  $dW$  is the **scalar product** of  $\vec{F}$  and the path  $d\vec{s}$   

$$dW = \vec{F} \cdot d\vec{s} \quad [W] = \text{J} = \text{W s} = \text{V A s} = \text{N m}$$
- In an **electric field** around a charge the force  $\vec{F}$  and the path  $d\vec{s}$  are either colinear or orthogonal (see figure on the right).
  - ⌈ Moving perpendicular to the field requires no work:  $dW_{\perp} = 0$
  - ⌈ Moving a **positive charge**  $q_{\text{test}}$  in field direction  $d\vec{s} = d\vec{r}$  **against the coulomb force**  $\|\vec{F}\| = -\|\vec{F}_C\|$  simplifies to:  

$$dW_{\parallel} = -\|\vec{F}_C\| dr = -q_{\text{test}}\|\vec{E}\| dr$$
- The total work  $W$  to move the charge from point  $a$  to  $b$  is then:

$$\frac{W}{q_{\text{test}}} = - \int_a^b \|\vec{E}\| dr = - \frac{Q}{4 \pi \varepsilon} \int_a^b \frac{dr}{r^2} = \frac{Q}{4 \pi \varepsilon} \left( \frac{1}{b} - \frac{1}{a} \right) = \varphi(b) - \varphi(a)$$

- **Electric potential**  $\varphi(r)$  is a scalar quantity and defines the work required to move a charge from infinity ( $a \rightarrow \infty$ ) to a distance  $r$  from the charge.

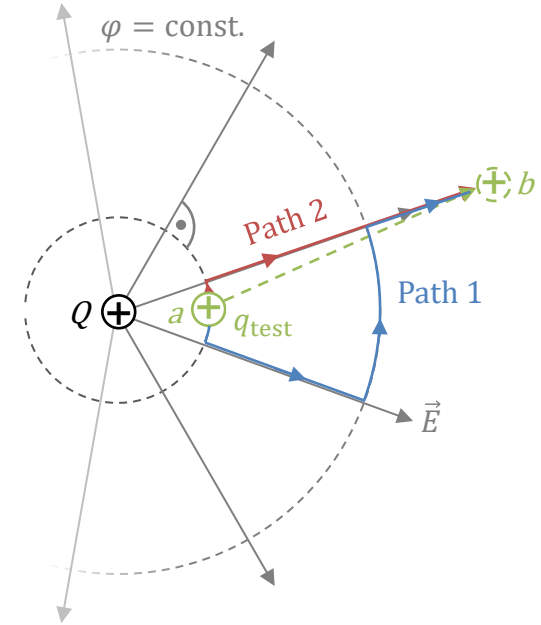
$$\varphi(r) = \frac{Q}{4 \pi \varepsilon r}$$

$$[\varphi] = \text{V} = \frac{\text{W}}{\text{A}} = \frac{\text{N m}}{\text{A s}}$$



$W < 0$  the test charge loses potential energy  
 $W > 0$  the test charge gains potential energy  
 Sign change for either a negative test charge or source charge!

VS

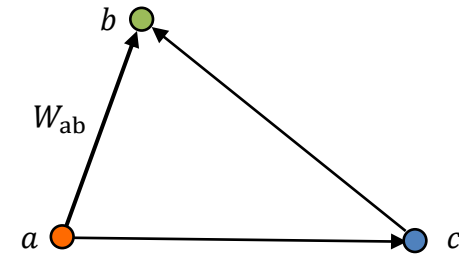
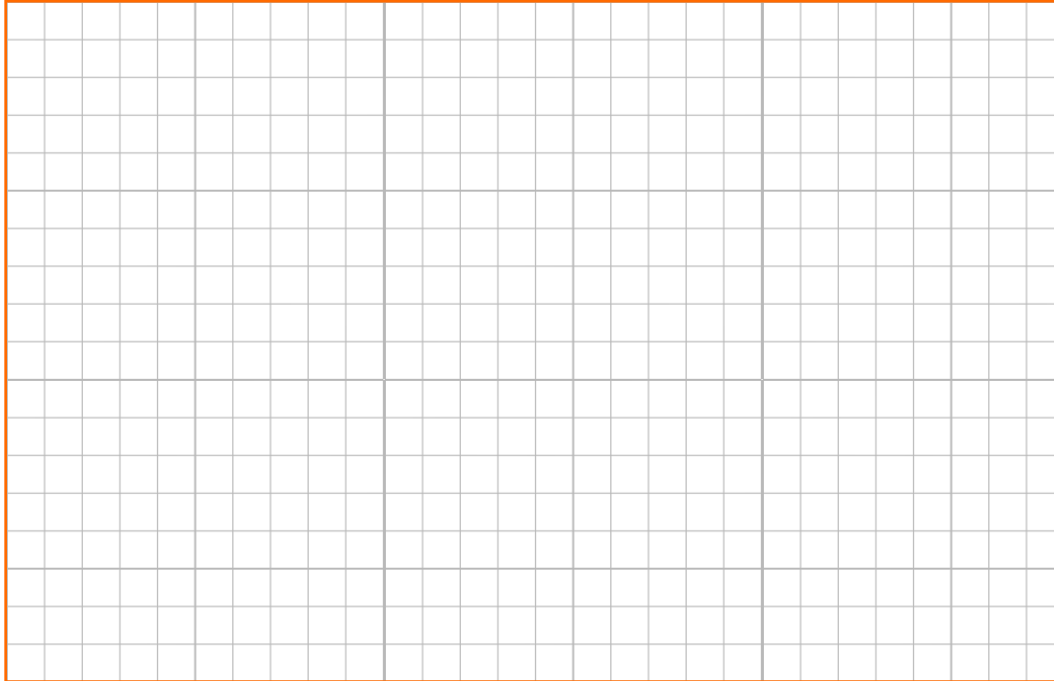


Work to move a charge from  $a$  to  $b$  in a coulomb field is independent of the path.

## 2.2 Basic concepts of electric fields

### Exercise: Path independence of work 1/2

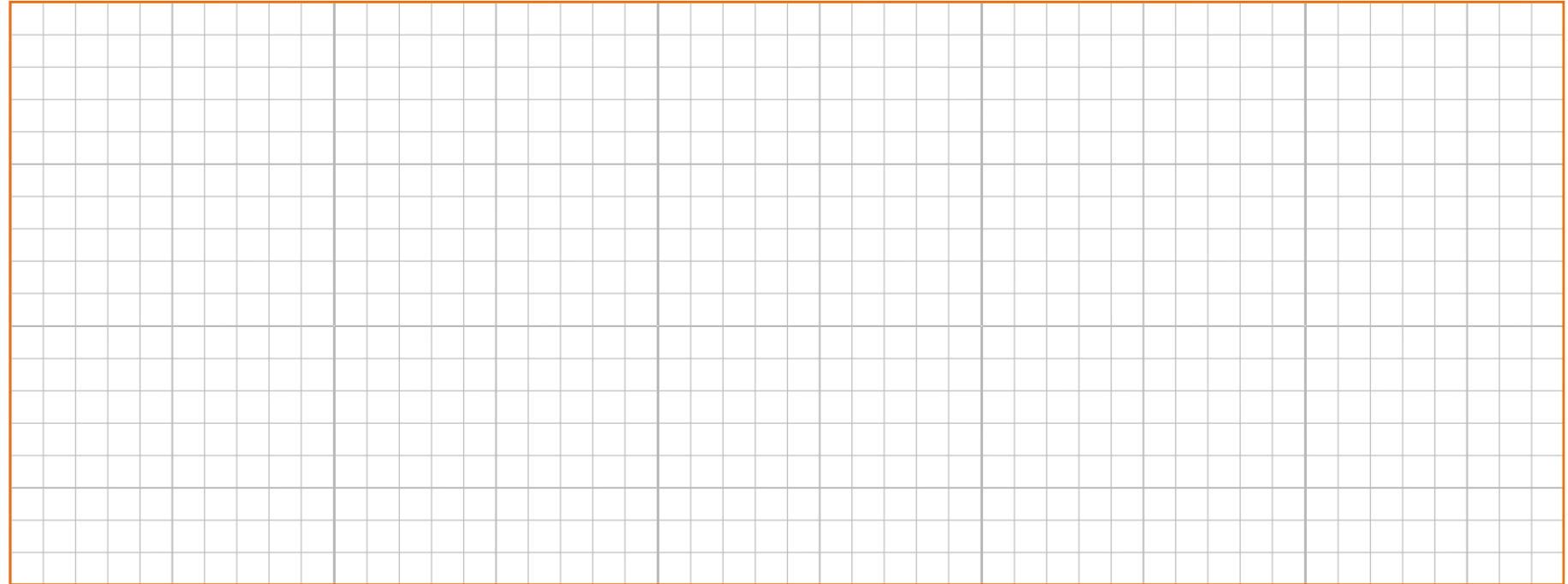
Show that the work  $W_{ab}$  to move a charged particle from  $a$  to  $b$  in an electric field  $\vec{E}$  is independent of the path  $a \rightarrow b$  by analysing an alternative path over  $c$ .



## 2.2 Basic concepts of electric fields

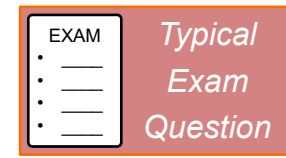
### Homework (advanced): Path independence of work 2/2

Show that the work to move a charged particle from  $a$  to  $b$  in an electric field  $\vec{F}$  is independent of the path  $a \rightarrow b$  by applying Maxwell's 2<sup>nd</sup> law  $\vec{\nabla} \times \vec{E} = -\vec{\nabla} \times (\vec{\nabla} \varphi) = 0$  Hint: using polar coordinates and check chapter 1.

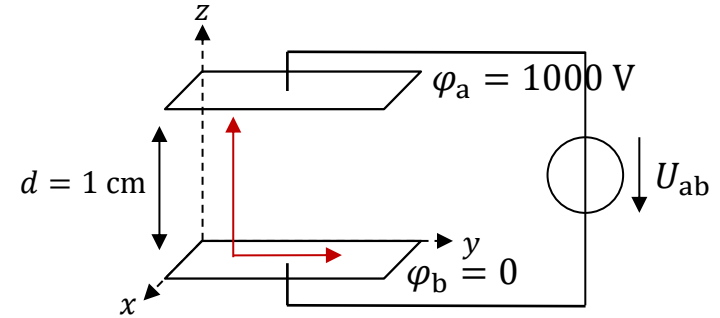
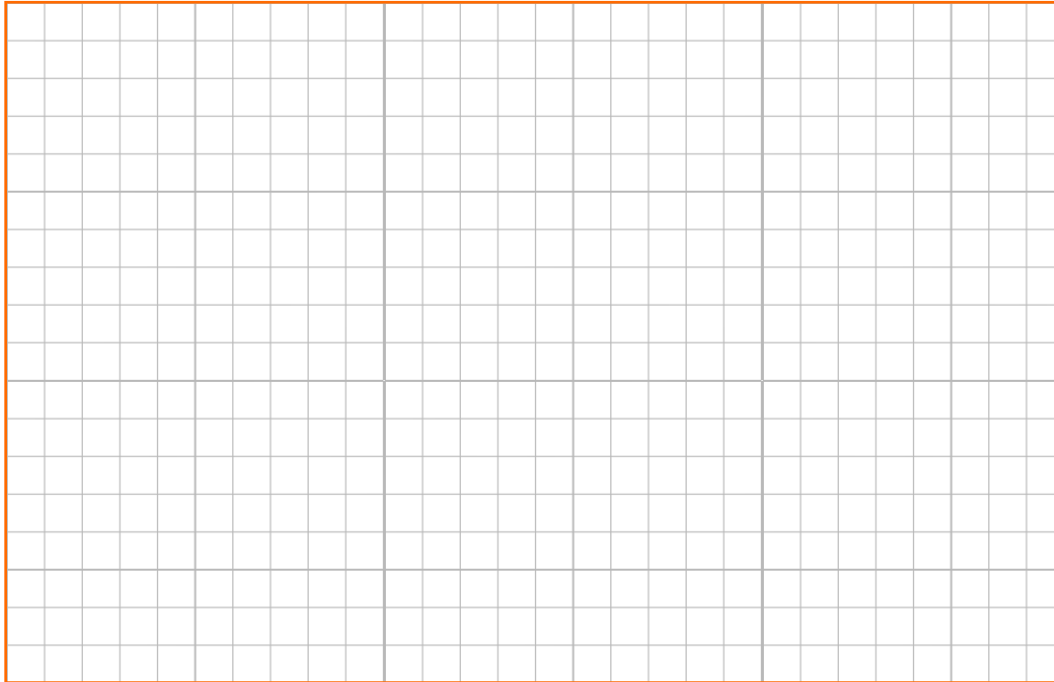


## 2.2 Basic concepts of electric fields

### Homework: Electric work



Calculate the work done to move a charge  $Q = 1 \text{ mC}$  from the positive  $\varphi_a$  to the negative  $\varphi_b$  plate in a parallel plate capacitor. What happens if the movement is reversed? What for lateral movement along the plate?



## 2.2 Basic concepts of electric fields

### Maxwell's 1<sup>st</sup> equation / Gauss' law

- Now that the electric potential is known, **revisit the concept of the electric field.**
- The **electric field** always points in the direction of the **steepest decline** of the **potential field**.
- This is the **definition of the gradient** (see chapter 1)
- Applying the del-operator to both sides results in Maxwell's 1st equation / Gauss' law in differential form

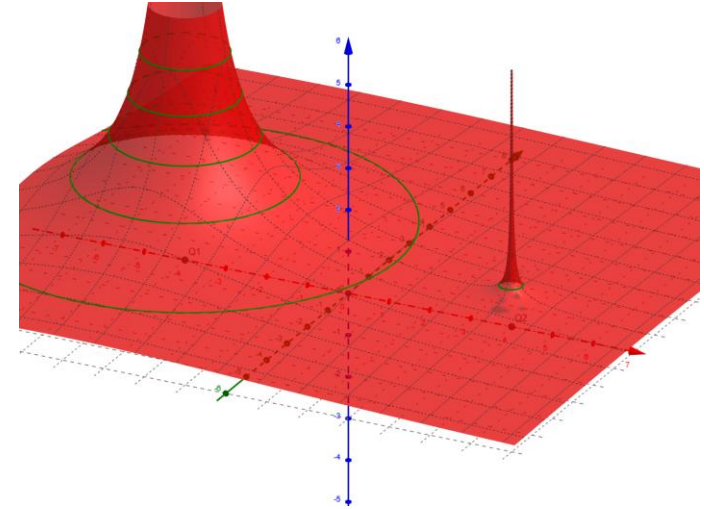
$$\vec{E} = -\vec{\nabla} \varphi$$

$$\vec{\nabla} \cdot \vec{E} = \vec{\nabla} \cdot (-\vec{\nabla} \varphi) = -\vec{\nabla}^2 \varphi = -\Delta \varphi = \frac{\rho}{\epsilon_0}$$

Where  $\vec{\nabla}^2 = \Delta$  is called the Laplace operator

*“A charge is the source of the electric field. The divergence of the field is equal to the charge density at that point.”*

–Differential form of Maxwell's 1<sup>st</sup> equation / Gauss' law



Equipotential lines of two charges (**top**) and 3D model (**bottom**).

## 2.2 Basic concepts of electric fields

### Example: Coulomb field

Calculate the divergence of the electric field  $\vec{\nabla} \cdot \vec{E}$  around a point charge  $Q$  using spherical coordinates. *Note:*

*The field in spherical coordinates is defined by  $\vec{\nabla} \cdot \vec{E} = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 E_r) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (\sin \theta E_\theta) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \phi} E_\phi$ .*



*Point charge  $Q$  generating a point symmetrical potential and an electric field.*

## 2.2 Basic concepts of electric fields

### Maxwell's 2<sup>nd</sup> equation

- Because a (static) **electric field** points in the direction of the steepest incline in a straight line, it does **not rotate**.
- Proof by applying the curl-operation to the electric field :

$$\vec{\nabla} \times \vec{E} = -\vec{\nabla} \times \vec{\nabla} \varphi = -\text{rot}(\text{grad } \varphi) = 0$$

See Appendix to 1.2.2:  
The del operator

*“Electric fields (source fields) are conservative and do not rotate.”*

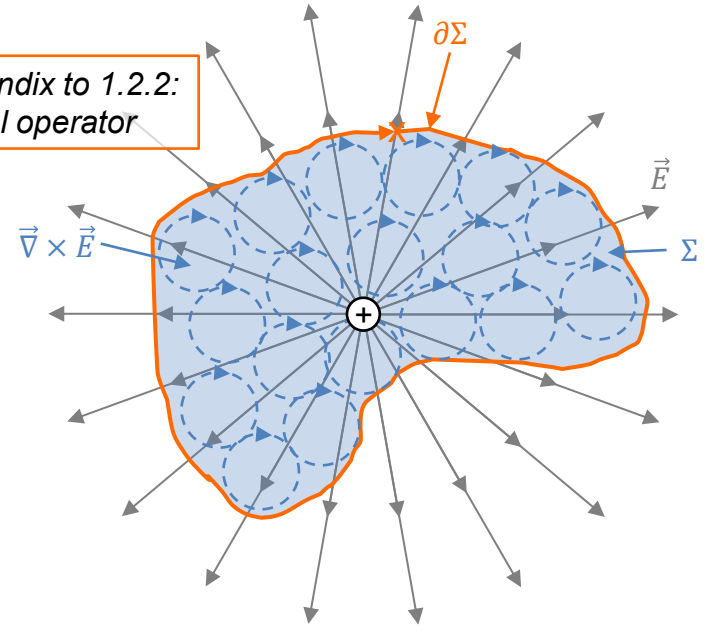
– Maxwell's 2<sup>nd</sup> equation / Stokes' theorem

- Applying the integral form of Stokes' theorem explains why the work  $W$  around a closed path  $l$  is zero for any path:

$$\underbrace{\oint_{\partial \Sigma} \vec{E} \cdot d\vec{l}}_{\text{work around closed path}} = \iint_{\Sigma} \underbrace{(\vec{\nabla} \times \vec{E})}_{\text{curl of } \vec{E} \text{ in area } a} \cdot d\vec{A} = 0$$



Works in the other direction as well: If the work is zero when moving a charge into a field, the field must be conservative.



*In an electric field, the work around a closed curve  $\partial \Sigma$  around a surface  $\Sigma$  is zero.*

## 2.2 Basic concepts of electric fields

### Voltage revisited

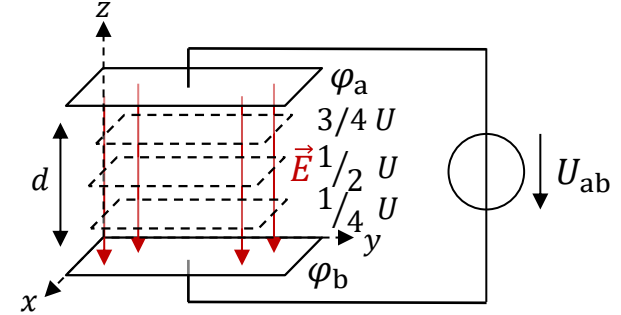
- Voltage  $U_{ab}$  is the **potential difference** or **work done to move a unit of charge**  $q_{\text{test}}$  from point  $a$  to point  $b$

$$U_{ab} = -\frac{W}{q_{\text{test}}} = \int_a^b \frac{\|\vec{F}\|}{q_{\text{test}}} dr = \int_a^b \vec{E} \cdot d\vec{s} = \varphi(a) - \varphi(b)$$

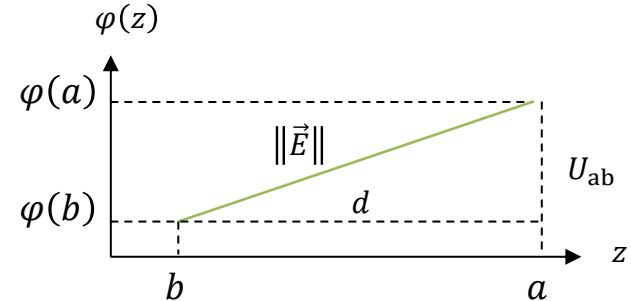
$[U] = \text{V}$

- If the **charge is positive**, **energy is released** ( $W < 0$ ) if charge moves from the **positive to the negative potential**.
- If the **charge is negative**, **energy is absorbed** ( $W > 0$ ) if charge moves from the **positive to the negative potential**.
- If the **field lines are parallel** (parallel plate capacitor) the line integral simplifies to the magnitude of the electric field  $\|\vec{E}\|$  and the distance  $d$  between.

$$U_{ab} = \|\vec{E}\| d$$



Schematic of a parallel plate capacitor.



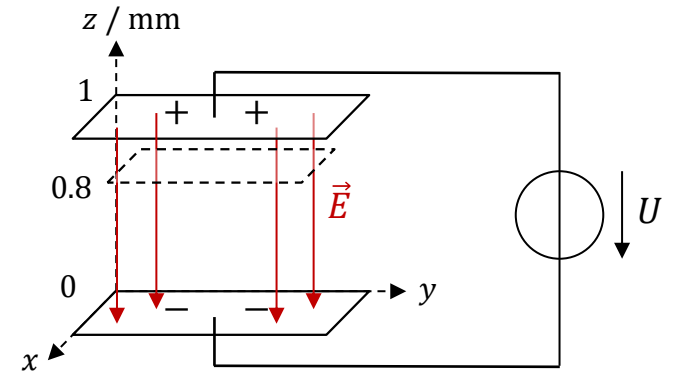
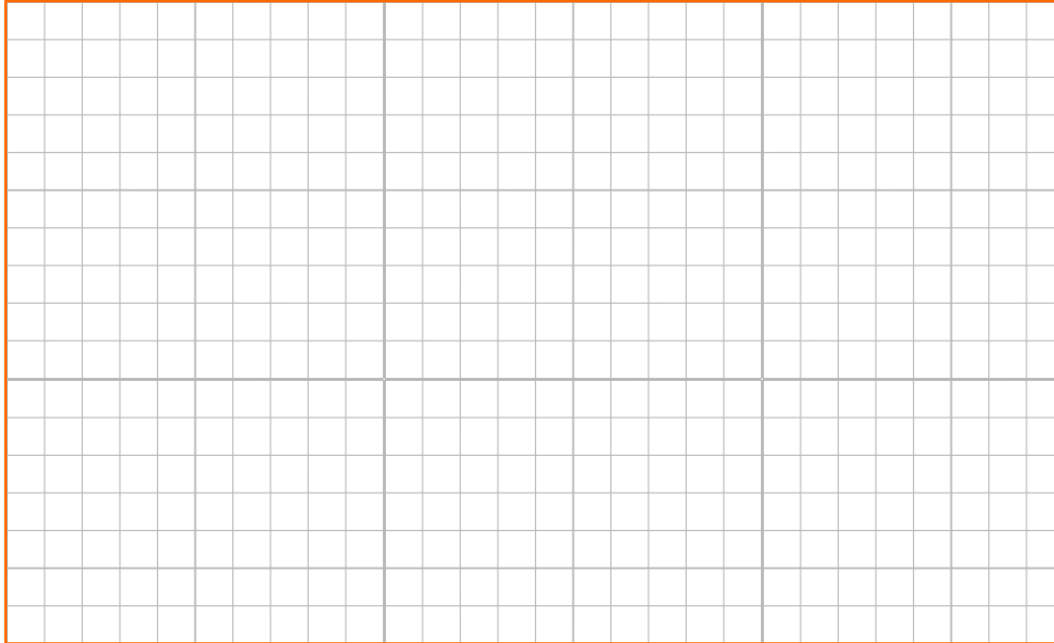
Linear voltage decrease within a parallel plate capacitor between plates  $a$  and  $b$ .

## 2.2 Basic concepts of electric fields

### Example: Electric field of a parallel plate capacitor

|      |          |
|------|----------|
| EXAM | Typical  |
| • —  | Exam     |
| • —  | Question |
| • —  |          |

Calculate the magnitude of the electric field  $\|\vec{E}\|$  for a parallel plate capacitor in vacuum. Both plates are in the  $xy$ -plane, the positive plate is located at  $z_+ = 1$  mm and the negative plate at  $z_- = 0$  mm. The voltage is  $U = 100$  V. What is the potential at  $z = 0.8$  mm?



## Summary

**Fundamental Quantities:** Charge  $Q$  (quantized, conserved), charge densities  $\lambda, \sigma, \rho$

**Forces & Fields:** Coulomb force  $\vec{F}_{12} = \frac{Q_1 Q_2}{4 \pi \epsilon r^2} \vec{e}_r$ ; Electric field  $\vec{E} = \vec{F}/q_{\text{test}}$ ; Superposition applies

**Energy & Potential:** Potential  $\varphi(r) = \frac{Q}{4 \pi \epsilon r}$ ; Voltage  $U_{ab} = \varphi_a - \varphi_b = \int \vec{E} \cdot d\vec{s}$ ; Work  $W_{ab} = Q(\varphi_b - \varphi_a)$  is path-independent

**Key Relationships:**  $\vec{E} = -\vec{\nabla} \varphi$  (gradient),  $\vec{\nabla} \times \vec{E} = 0$  (conservative),  $\vec{\nabla} \cdot \vec{E} = \frac{\rho}{\epsilon_0}$  (Gauss' law)

## 2.2 Basic concepts of electric fields

## Learning objectives

Students will be able to:

Define  $\vec{D} = \varepsilon_0 \varepsilon_r \vec{E}$  and explain role of permittivity  $\varepsilon$

Differentiate between polarization (dielectrics: bound charge alignment) and induction (conductors: mobile charge redistribution)

Explain how relative permittivity  $\varepsilon_r$  affects electric field  $\vec{E}$ , capacitance  $C$ , potential  $\varphi$ , and voltage  $U$  while keeping  $\vec{D}$  constant

## 2.3 Materials

## 2.3 Materials

### Electric flux density and polarization

- In dielectrics, bound charges shift slightly in an applied field → **polarization**
- Polarization creates an **internal field** opposing the applied field → total  $\vec{E}$  inside the medium is reduced
- Electric flux density  $\vec{D}$  (also displacement field) accounts for **free charges only** and is **independent of the medium**
- The relation between  $\vec{D}$  and  $\vec{E}$  is
  - ⌈ Vacuum

$$\vec{D} = \epsilon_0 \vec{E} \quad [D] = \frac{\text{A s}}{\text{m}^2}$$

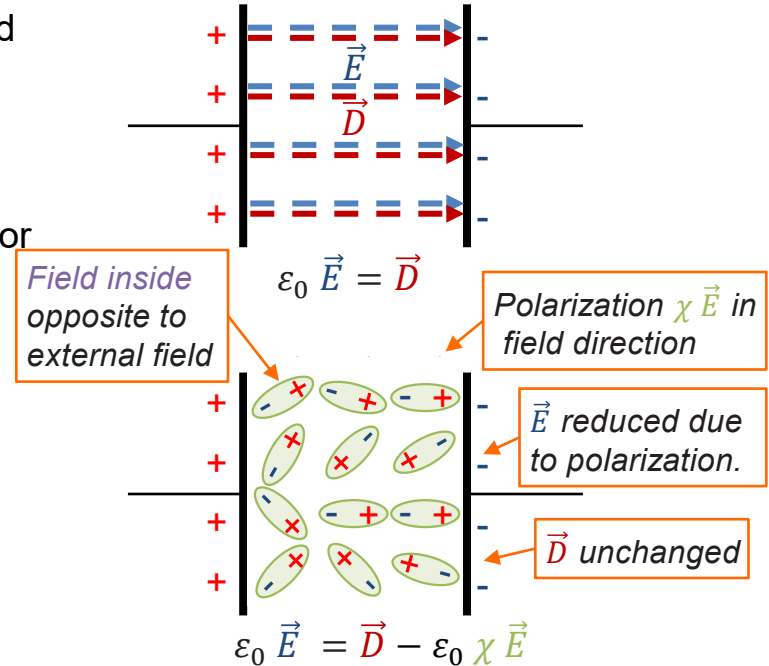
- ⌈ Linear, isotropic media

$$\vec{D} = \epsilon_0 \vec{E} + \underbrace{\epsilon_0 \chi \vec{E}}_{\text{polarization}} = \epsilon_0 (1 + \chi) \vec{E} = \epsilon_0 \epsilon_r \vec{E}$$

Where  $\chi$  is the electric **susceptibility** and the **relative permittivity**  $\epsilon_r = 1 + \chi$  describe the degree of polarization



Electric flux density is a property of the charge distribution itself, whereas the electric field also reflects the influence of the surrounding medium



The electric displacement field between to parallel plates filled with vacuum (**top**) and a dielectric (**bottom**).

## 2.3 Materials

### About dielectrics

- Many vacuum electrostatics results carry over to a **linear, homogeneous, isotropic** material by replacing  $\varepsilon_0$  with  $\varepsilon = \varepsilon_0 \varepsilon_r$ .
- Adding a **dielectric**  $\varepsilon_r$  **reduces** the
  - ⌋ Electric field  $\vec{E}$
  - ⌋ Coulombic force  $\vec{F}$
  - ⌋ Electric potential  $\varphi$
  - ⌋ Voltage  $U$
  - ⌋ Work  $W$
- Adding a **dielectric**  $\varepsilon_r$  **increases** the
  - ⌋ Capacitance  $C$
- A **dielectric has no effect** on the
  - ⌋ Charge  $Q$
  - ⌋ Electric flux density  $\vec{D}$

| Dielectric material | Relative permittivity $\varepsilon_r$ | Comment                       |
|---------------------|---------------------------------------|-------------------------------|
| Vacuum              | 1                                     | no dipoles                    |
| Air                 | 1.0006                                | Low mass, non-polar molecules |
| Mineral oil         | 2.2                                   | little polarity               |
| Water               | 80                                    | High polarity of $H_2O$       |
| Polypropylene       | 2.3                                   |                               |
| Paper               | 2.3 – 4                               | Used in cheap capacitors      |
| Porcelain           | 5 – 7                                 | Ceramic capacitors            |
| Special ceramics    | 100 – 400                             |                               |



Quick  
check

1. Explain in your own words why a dielectric reduces the  $\vec{E}$
2. Why does a dielectric increase  $C$ ?
3. Why does it not have influence on  $\vec{D}$ ?

## 2.3 Materials

### Exercise: Field around a point charge

|         |          |
|---------|----------|
| EXAM    | Typical  |
| • _____ | Exam     |
| • _____ | Question |
| • _____ |          |
| • _____ |          |

Draw the field lines and equipotential surfaces around a positive point charge  $Q$  at a distance  $r$  in vacuum ( $\epsilon_r = 1$ ). Calculate  $\varphi$ ,  $\vec{E}_{\text{vac}}$  and  $\vec{D}_{\text{vac}}$



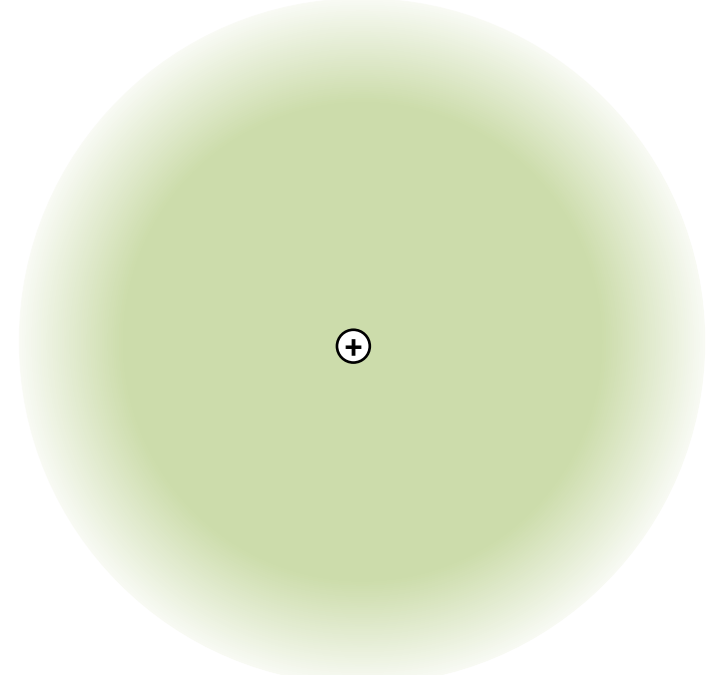
*Point charge  $Q$  generating a point symmetrical potential and an electric field.*

## 2.3 Materials

### Homework: Field around a point charge in a medium

|      |                             |
|------|-----------------------------|
| EXAM | Typical<br>Exam<br>Question |
| •    |                             |
| •    |                             |
| •    |                             |

Now reassess the situation when the charge is placed in a medium with  $\epsilon_r = 2$ . Compare to before.

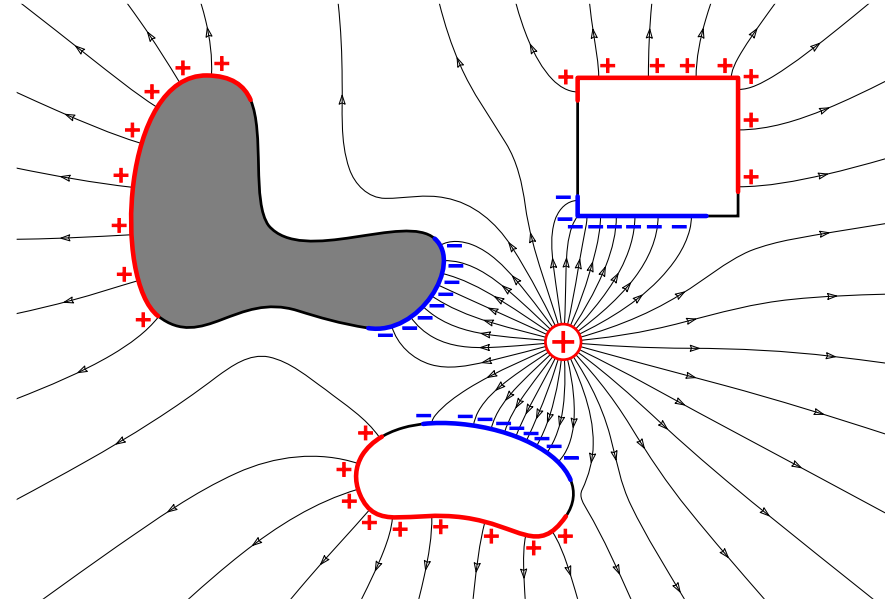


*Point charge  $Q$  generating a point symmetrical potential and an electric field.*

## 2.3 Materials

### Electrostatic induction

- Electrostatic induction (influence) is the **redistribution** of electric charge in a metallic object **due to** repulsive or attractive force of **nearby charges**.
- Induced **charges** reside **on the surface** of conducting objects and generate an electric field opposite to the external field.
- The electric **fields inside is zero** (“Faraday’s cage”)
- *Exercise: Prove, why the there is no electric field inside metallic objects is always zero. Recall that metallic spheres are equipotential surfaces.*



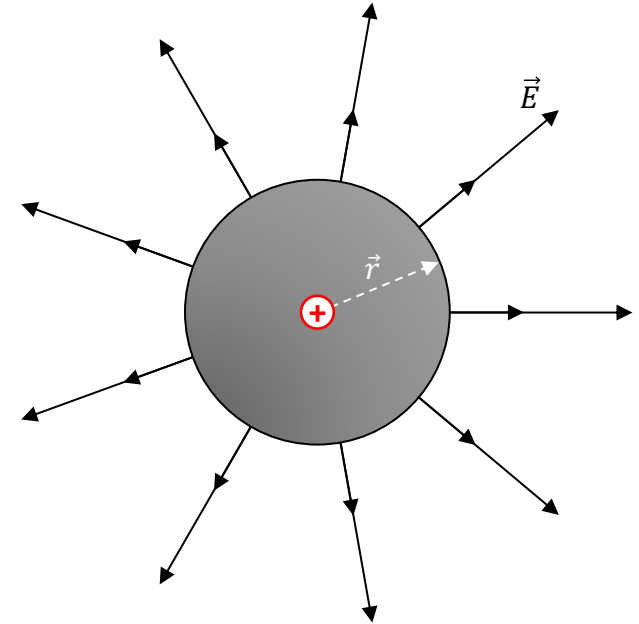
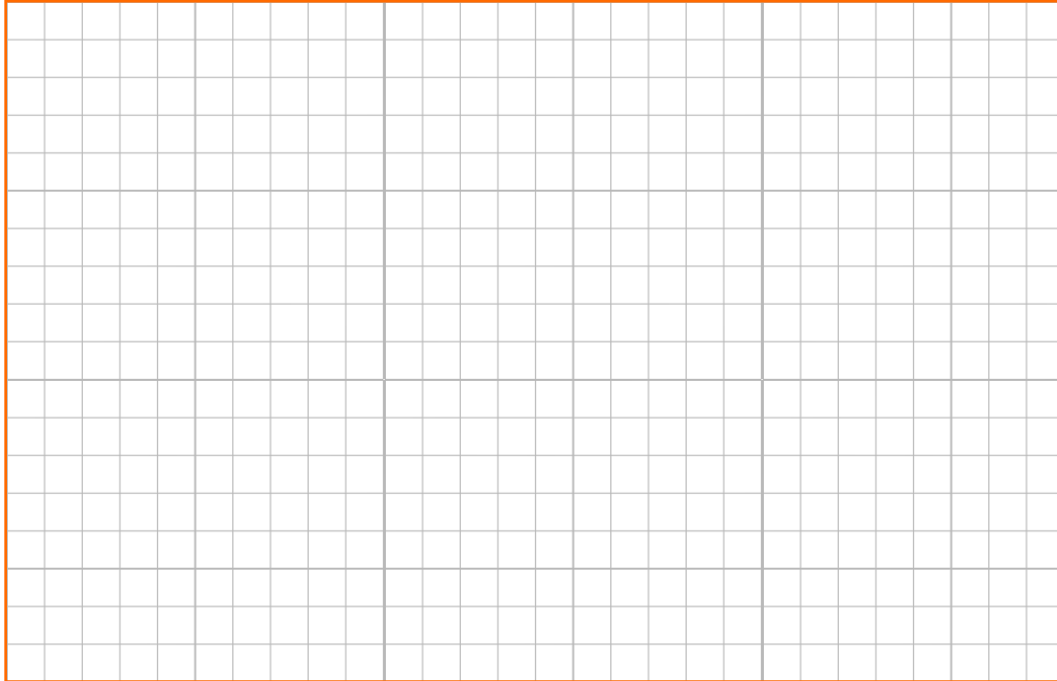
*Surface charges induced in metal objects by a nearby charge.*

## 2.3 Materials

### Example: Field around a point charge in a medium

|      |          |
|------|----------|
| EXAM | Typical  |
| •    | Exam     |
| •    | Question |
| •    |          |
| •    |          |

Now reconsider the situation where a positive point charge  $+Q$  is added at the centre of a neutral, conducting sphere of radius  $R$ . Compare this to the case without the conductor.



*Simplified visualization of a point charge in a metallic sphere*

## Summary

**Polarization vs. induction:** Bound charges align (dielectrics) vs. mobile charges redistribute (conductors)

**Electric flux density:**  $\vec{D} = \varepsilon_0 \varepsilon_r \vec{E}$  (medium-independent external field);

**Dielectric effects:** Reduces  $\vec{E}, \varphi, U, W$  by factor  $\varepsilon_r$ ; Increases  $C$  by  $\varepsilon_r$ ; No effect on  $Q, \vec{D}$

## 2.3 Materials

## Learning objective

Students will be able to:

- Define electric flux ( $\Phi = \oint \vec{D} \cdot d\vec{A} = \int \vec{\nabla} \cdot \vec{D} dV$ ) and apply Gauss' law ( $\oint \vec{D} \cdot d\vec{A} = Q_{\text{enclosed}}, \vec{\nabla} \cdot \vec{D} = \rho$ )
- Construct Gaussian surfaces for symmetric problems (spherical, cylindrical, planar) and **calculate**  $\|\vec{E}\|$  and  $\|\vec{D}\|$  without complex integration  $\pm \|\vec{D}\| A = Q_{\text{inside}}$ ; Calculate  $\vec{E}$  and  $\vec{D}$  by exploiting the geometry.
- Analyse electrostatic induction: explain why  $\vec{E} = 0$  inside conductors and **calculate** surface charge densities

## 2.4 Electric flux and Gauss' law

## 2.4 Electric flux and Gauss' law

### Gauss' law

- Recall the differential form of Gauss' law (chapters 2.1 & 2.2)

$$\vec{\nabla} \cdot \vec{D} = \rho$$

- Using the definition of the **flux** from chapter 1.2.2, the integral form of Gauss' law can be formulated:

$$\Phi = \oiint_{\partial\Omega} \vec{D} \cdot d\vec{A} = \iiint_{\Omega} \vec{\nabla} \cdot \vec{D} dV = \iiint_{\Omega} \rho dV = \begin{cases} Q & Q \text{ inside } \partial\Omega \\ 0 & Q \text{ outside } \partial\Omega \end{cases}$$

- The unit of the flux is

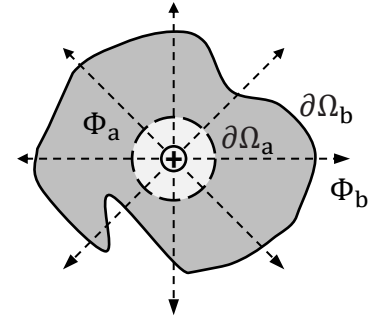
$$[\Phi] = \text{A s}$$

*“The **flux**  $\Phi$  through a closed surface is **independent of the surface area and shape** and equal to the **charge enclosed by the surface**.”*

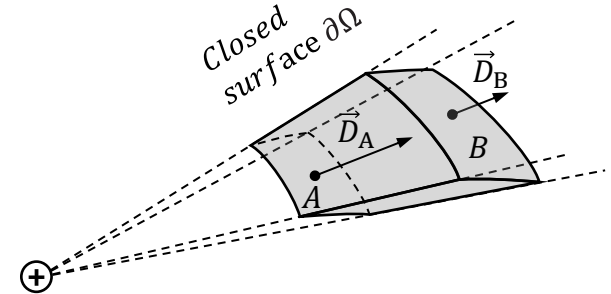
–Integral form of Maxwell's 1<sup>st</sup> equation / Gauss' law

- For systems where the field  $\vec{D}$  is constant and perpendicular to the area  $\vec{A}$  enclosing  $Q_{\text{inside}}$  lies within, Gauss' law simplifies

$$\Phi = \pm \|\vec{D}\| A = Q_{\text{inside}}$$



*The flux through surface  $A_a$  is identical to the flux through surface  $A_b$  and proportional to the charge within.*

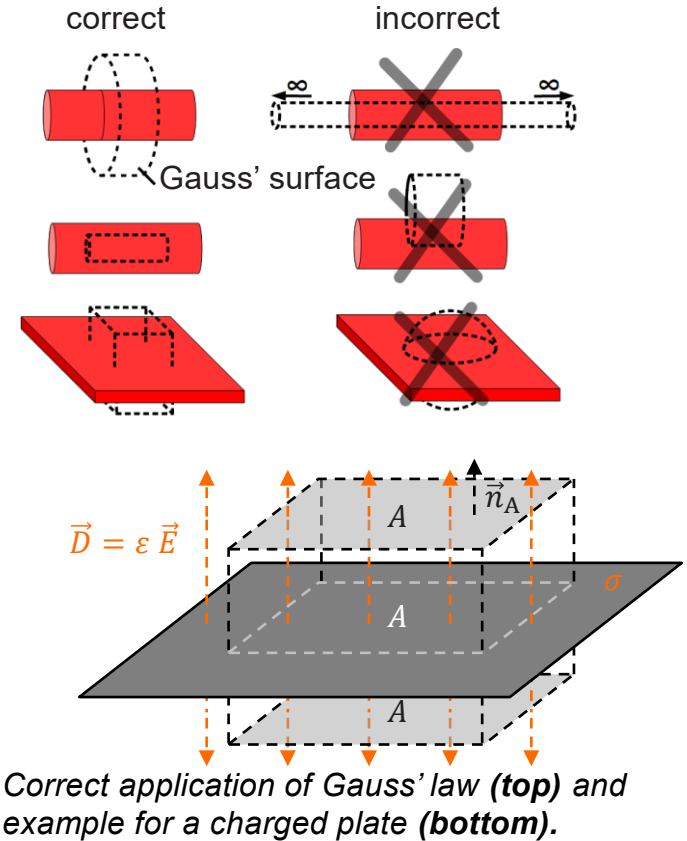


*The flux through a surface  $\partial\Omega$  is zero if the charge is outside of the surface.*

## 2.4 Electric flux and Gauss' law

### Application of Gauss' law

- **Gauss' surface:** imaginary closed surface where  $\|\vec{D}\|$  is the same everywhere.
- **Step-by-step approach:**
  - 1 **Choose the shape** so that  $\|\vec{D}\|$  is constant: sphere (point charge), cylinder (line charge), box (plane).
  - 1 **Centre** the Gauss' surface on the charge.
  - 1 **Compute**  $\|\vec{D}\|$  from the enclosed charge, vice versa
- *Exercise: Calculate the magnitude of the electric displacement field  $\|\vec{D}\|$  and electric field  $\|\vec{E}\|$  of an infinite plane with a charge density  $\sigma$  as in the image to the right.*

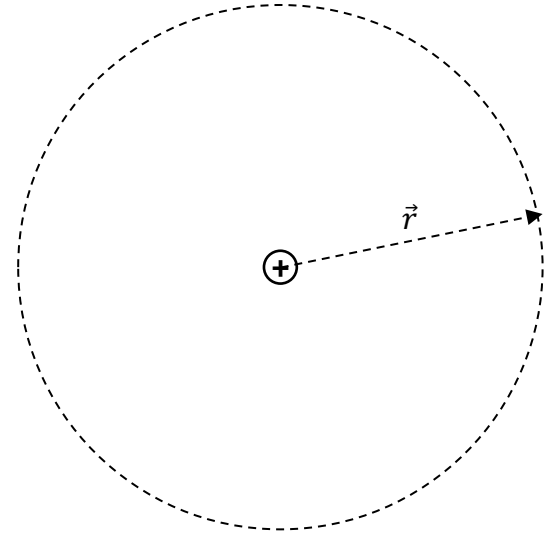
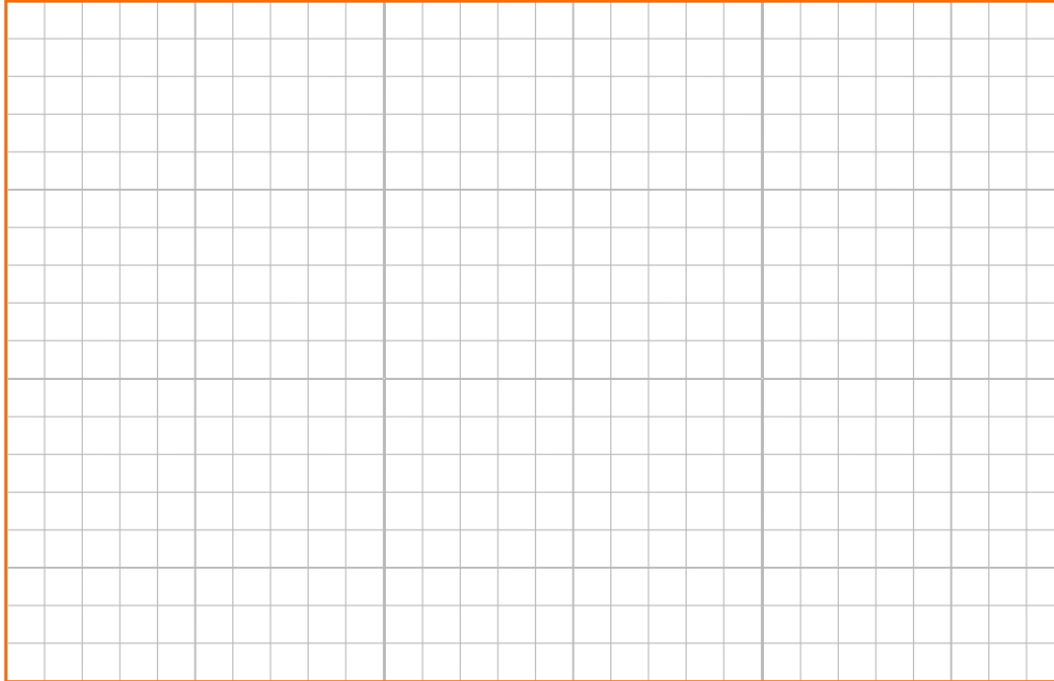


## 2.4 Electric flux and Gauss' law

Exercise: Field around a point charge by applying a Gauss' surface

|      |          |
|------|----------|
| EXAM | Typical  |
| • —  | Exam     |
| • —  | Question |
| • —  |          |

Calculate  $\|\vec{D}\|$  around a positive point charge  $Q$  at a distance  $r$  in vacuum ( $\epsilon_r = 1$ ) using Gauss formalism.  
Compare to the previous exercises



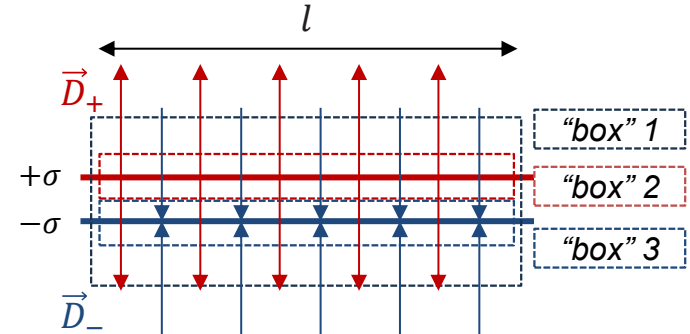
*Point charge  $Q$  generating a point symmetrical potential and an electric field.*

## 2.4 Electric flux and Gauss' law

### Example: Application of Gauss' law

|      |          |
|------|----------|
| EXAM | Typical  |
| •    | Exam     |
| •    | Question |
| •    |          |
| •    |          |

Use the Gauss' methodology to calculate the electric field inside and outside of a parallel plate capacitor where the positive plate is charged with a charge density of  $+\sigma$  and the negative plate is charged with  $-\sigma$ .

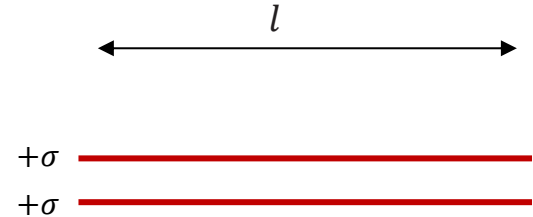


*Applying Gauss' law on an ideal parallel plate capacitor. Field lines leave the positive plate and enter the negative platen. Between the plates the field of both plates adds up. Outside the field lines cancel out each other.*

## 2.4 Electric flux and Gauss' law

### Homework: Application of Gauss' law

Use the Gauss's box methodology to calculate the electric field inside and outside of two positively charged plates, charged with a charge density of  $+\sigma$ .



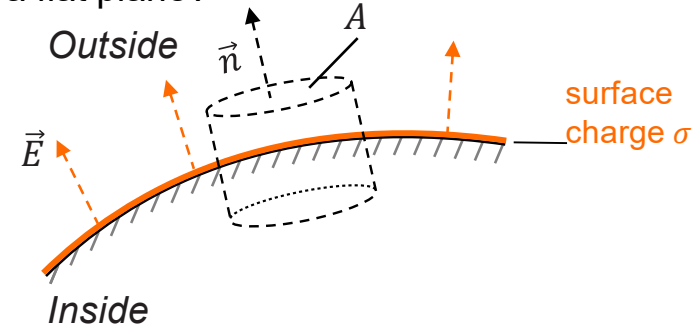
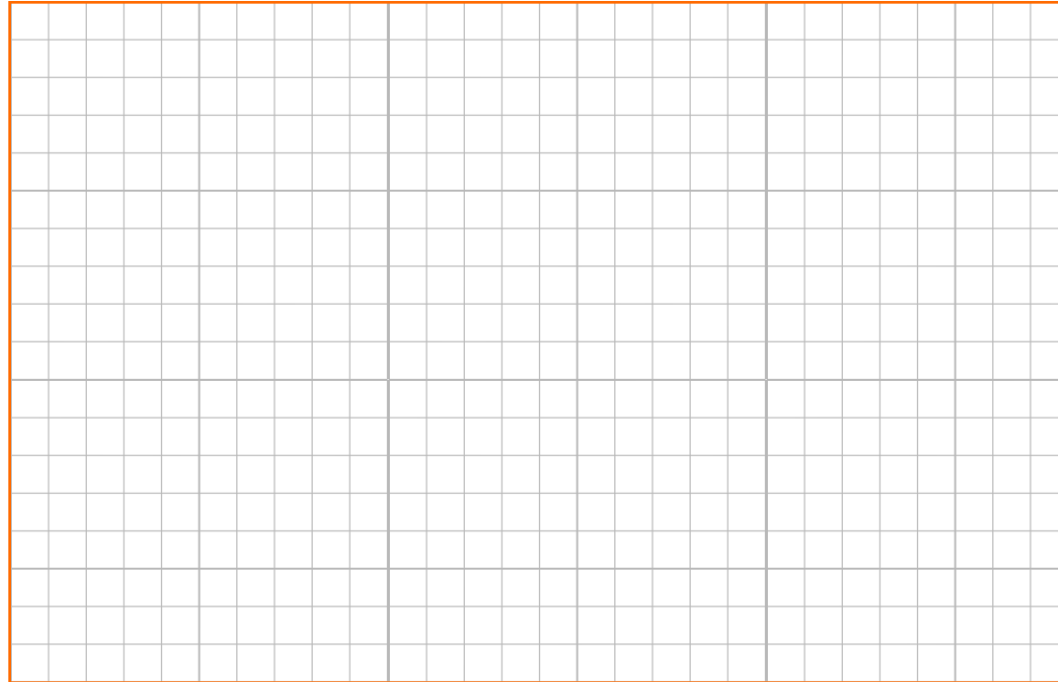
*Applying Gauss' law on two positively charged parallel plates.*

## 2.4 Electric flux and Gauss' law

### Example: Application of Gauss' law

|      |          |
|------|----------|
| EXAM | Typical  |
| •    | Exam     |
| •    | Question |

Use Gauss' law to probe the surface charge density  $\sigma$  of a sphere for a given electric field  $\vec{E}$ . How will the magnitude of the electric field change, when the sphere is opened to a flat plane?



*A tiny Gauss's box whose sides are perpendicular to a conductor's surface is used to find the local surface charge once the electric potential and the electric field are calculated by solving Laplace's equation. The electric field is perpendicular, locally, to the equipotential surface of the conductor, and zero inside; its flux  $\pi a^2 \cdot E$ , by Gauss's law equals  $\pi a^2 \cdot \sigma / \epsilon_0$ . Thus,  $\sigma = \epsilon_0 E$ .*

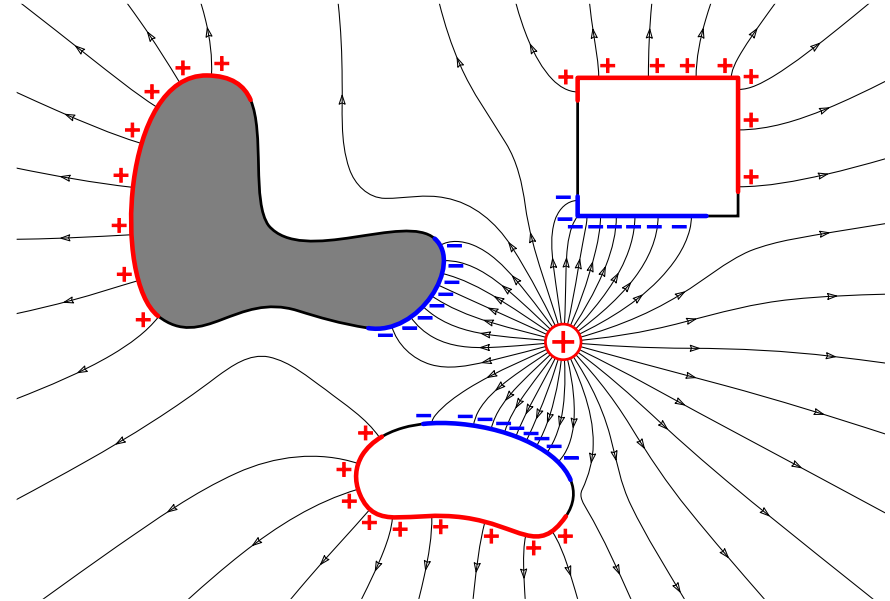
## 2.4 Electric flux and Gauss' law

Electrostatic induction derived from Gauss law.

- Electrostatic induction can now also be understood in terms of Gauss' law.
- Because there are no charges inside, the total flux inside of metallic objects is zero.
- The electric **fields inside is zero**.
- *Exercise: Prove, why the field inside metallic objects is always zero applying Gauss' law.*

? Discuss potential locations/shapes for a Gauss' box? Why can it only be placed around the point charge?

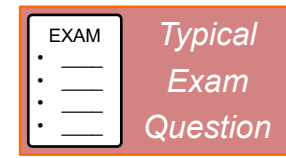
Quick check



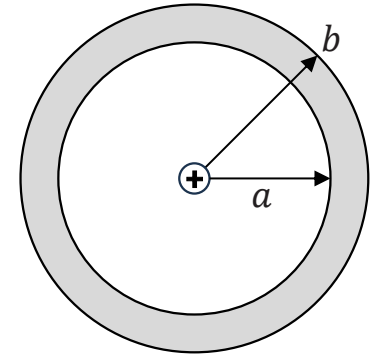
*Surface charges induced in metal objects by a nearby charge.*

## 2.4 Electric flux and Gauss' law

### Example: Electrostatic induction



Imagine a point charge  $Q$  inside of a metallic sphere in vacuum with inner diameter  $a$  and an outer diameter  $b$ . Calculate the electric field  $\vec{E}$  at a distance a)  $r < a$ , b)  $a < r < b$  and c)  $r > b$  and draw the field lines.



*Schematic of a point charge inside a metallic sphere in vacuum.*

## Summary

**Electric Flux:**  $\Phi = \oint \vec{D} \cdot d\vec{A} = \int \vec{\nabla} \cdot \vec{D} dV$ , represents field ‘flow’ through surface

**Gauss’ Law:** Integral:  $\oint \vec{D} \cdot d\vec{A} = Q_{\text{enclosed}}$ ; Differential:  $\vec{\nabla} \cdot \vec{D} = \rho$  (Maxwell’s 1st equation)

**Gauss Box Method:**

1. Identify symmetry: sphere (point charge), cylinder (line charge), box (plane).
2. Choose surface where  $\vec{D}$  is constant
3. Apply  $\|\vec{D}\| A = Q_{\text{enclosed}}$
4. Calculate  $\|\vec{E}\| = \frac{\|\vec{D}\|}{\epsilon}$  and the respective vector quantities  $\vec{E}, \vec{D}$  from the geometry if necessary

**Key Results:** Infinite plane  $\|\vec{E}\| = \frac{\sigma}{2\epsilon}$ ; Parallel plates  $\|\vec{E}\|_{\text{in}} = \frac{\sigma}{\epsilon}, \|\vec{E}\|_{\text{out}} = 0$ ; Point charge  $\|\vec{E}\| = \frac{Q}{4\pi\epsilon r^2}$

**Induction:** Conductors are equipotential ( $\varphi = \text{const.} \rightarrow \vec{E}_{\text{inside}} = 0$ ); Charges on surface; Faraday cage effect

## 2.4 Electric flux and Gauss’ law

## Learning objective

Students will be able to:

### – **Energy Storage:**

- ⌋ Calculate capacitance from geometry using Gauss' law  $\left(C = \frac{\varepsilon}{\int_{r_1}^{r_2} \frac{1}{A} dr}\right)$  and compute stored energy

$$\left(W = \frac{1}{2} C U^2, w_V = \frac{1}{2} \varepsilon \|\vec{E}\|^2\right)$$

- ⌋ Determine maximum voltage from dielectric strength  $(\|\vec{E}\| \leq \|\vec{E}_{\max}\|)$

### – **Conductivity:**

- ⌋ Explain current's microscopic origin  $(\vec{v}_d = \mu \vec{E}, \vec{j} = n |Q| \vec{v}_d = \kappa \vec{E})$  and **derive** Ohm's law  $\left(R = \frac{U}{I} = \rho \frac{l}{A}\right)$  from field theory
- ⌋ Calculate conductor properties  $(\mu, \vec{v}_d, \kappa, \rho)$  from material parameters

# 2.5 Field applications: energy storage and conductivity

## 2.5 Field applications: energy storage and conductivity thws

### Capacitance

- Any arrangement of **two electrodes** with an **insulating layer** in between is named **capacitor**.
- The **proportionally** between applied **voltage**  $U$  and stored **charge**  $Q$  is the **capacitance**  $C$ :

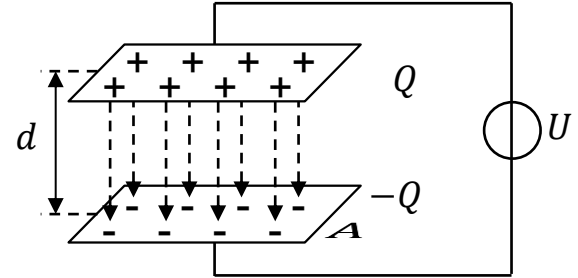
$$C = \frac{Q}{U} \quad [C] = \frac{\text{As}}{\text{V}} = \text{F}$$

- Using the voltage definition  $U = \int_{r_1}^{r_2} \|\vec{E}\| dr$  from earlier, the capacitance can be calculated for **any arbitrary shape**:

$$C = \frac{Q}{\int_{r_1}^{r_2} \|\vec{E}\| dr} = \frac{\varepsilon}{\int_{r_1}^{r_2} \frac{1}{A} dr}$$



As we'll see later, already a single electrode alone constitutes a capacitor.



Get  $\|\vec{E}\|$  from Gauss' law  $\|\vec{E}\| A = \frac{Q}{\varepsilon}$

- Special capacitor geometries** (constant  $\varepsilon = \varepsilon_0 \varepsilon_r$ !)

Parallel plates:  $U = \frac{Q}{\varepsilon A} (r_2 - r_1) = \frac{Q}{\varepsilon A} d \quad \therefore$

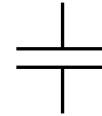
Spherical conductors:  $U = \frac{Q}{4 \pi \varepsilon} \left( \frac{1}{r_1} - \frac{1}{r_2} \right) \quad \therefore$

Cylindrical / coaxial cable:  $U = \frac{Q}{2 \pi \varepsilon l} \ln \frac{r_2}{r_1} \quad \therefore$

$$C = \varepsilon \frac{A}{d}$$

$$C = \frac{4 \pi \varepsilon}{\left( \frac{1}{r_1} - \frac{1}{r_2} \right)}$$

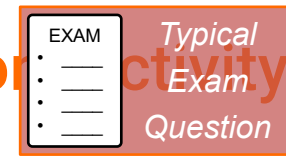
$$C = \frac{2 \pi \varepsilon l}{\ln \frac{r_2}{r_1}}$$



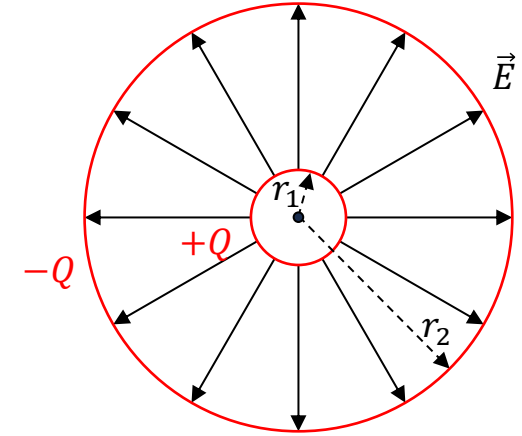
Schematic of a parallel plate capacitor (**top**).  
Symbol for a capacitor according to DIN EN-60617 (**bottom**).

## 2.5 Field applications: energy storage and capacitance

### Example: Spherical capacitor



Calculate the electric field inside and outside and the capacitance of a spherical capacitor in vacuum where the inner sphere of radius  $r_1$  is positively charged and the outer sphere of radius  $r_2$  is negatively charged. What happens to the voltage  $U$  and capacitance  $C$  if you increase the outer diameter towards infinity?

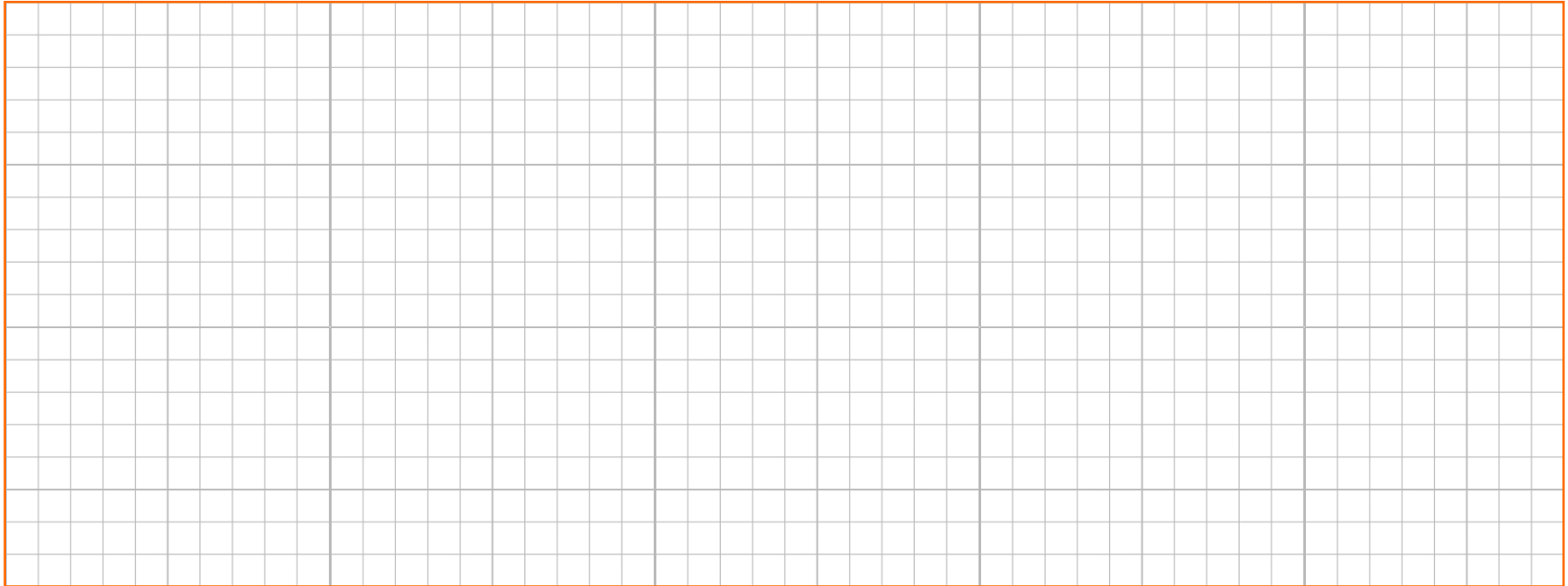


*Schematic of a spherical capacitor in vacuum.*

## 2.5 Field applications: energy storage and conductivity thws

Homework: Capacitance of a parallel plate capacitor.

Derive the capacitance of a parallel plate capacitor using the Gauss' formalism



## 2.5 Field applications: energy storage and conductivity thws

### Energy

- Capacitors **store electric energy** in the electric field.
- Modify the definition of the electric work from chapter 2.2 for a homogeneous field inside a parallel plate capacitor of 'infinite' area:

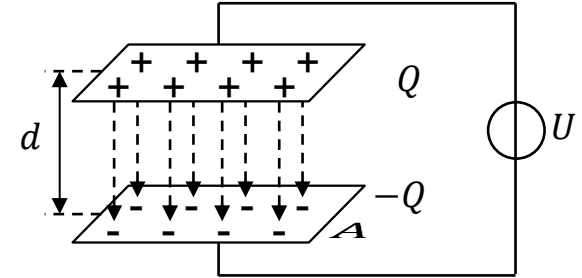
$$dW = U dQ = \frac{Q}{C} dQ$$

- Integrate from 0 to  $Q$  and substitute  $Q = C U$  to get the energy  $W$  of a capacitor charged to a voltage  $U$  :

$$W = \frac{1}{2} \frac{Q^2}{C} = \frac{1}{2} C U^2 = \frac{1}{2} Q U \quad [W] = \text{Wh}$$

- Substitute  $C = \varepsilon \frac{A}{d}$  and  $U = \|\vec{E}\| d$  to yield the volumetric energy density  $w_V$ , where  $V$  is the volume:

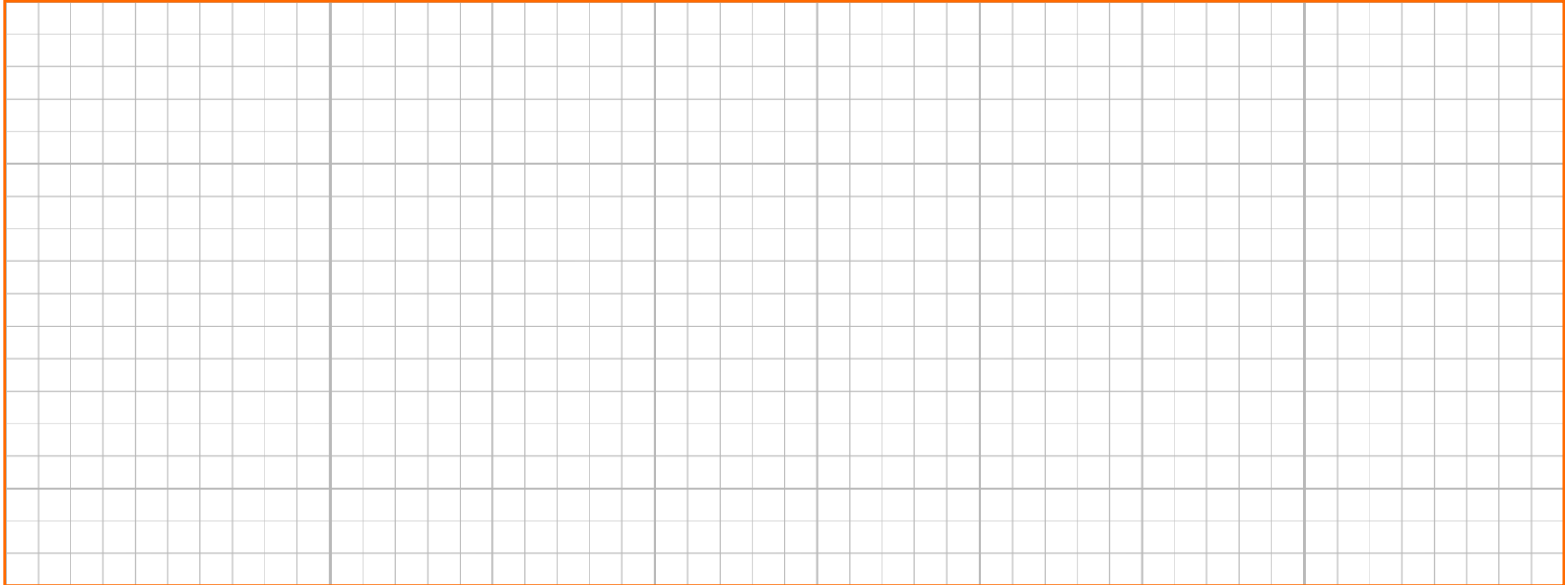
$$w_V = \frac{W}{V} = \frac{1}{2} \varepsilon \|\vec{E}\|^2 \quad [w_V] = \frac{\text{Wh}}{\text{m}^3}$$



## 2.5 Field applications: energy storage and conductivity thws

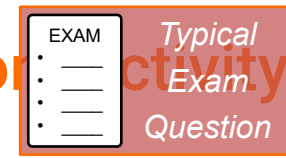
Homework: Energy stored in a parallel plate capacitor

Show that in a parallel plate capacitor the following equation is true:  $w_V = \frac{1}{2} C U^2 = \frac{1}{2} \|\vec{D}\| \|\vec{E}\|$ .

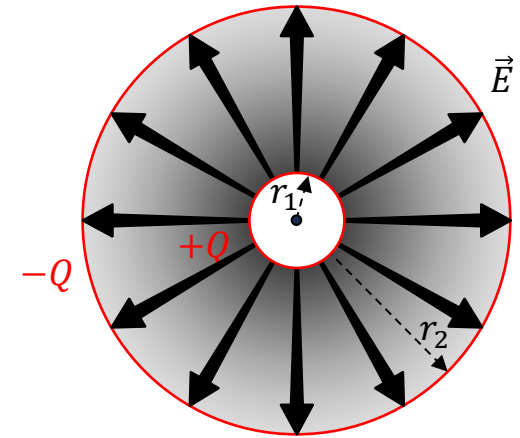


## 2.5 Field applications: energy storage and conductivity

### Example: Spherical capacitor with non-constant permittivity



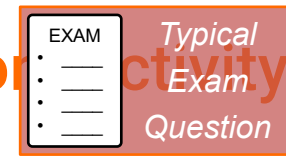
Calculate the electric field inside and outside and the capacitance of a spherical capacitor where the inner sphere of radius  $r_1$  is positively charged and the outer sphere of radius  $r_2$  is negatively charged. The dielectric shall be a quadratic function of the radius  $\varepsilon_r = \frac{\varepsilon_1}{r^2}$ . *Note: Use  $r = \|\vec{r}\|$  and  $E = \|\vec{E}\|$  for better readability.*



*Schematic of a spherical capacitor with changing dielectric. The arrow indicate a constant electric field.*

## 2.5 Field applications: energy storage and conductivity

### Homework: Cylindrical capacitor / Coaxial cable

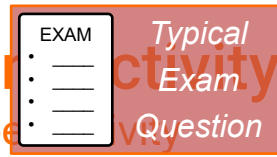


A coaxial cable consists of an inner conductor of radius  $r_1$  surrounded by a concentric conducting shield of radius  $r_2$ , with the two separated by a dielectric  $\epsilon_r$ . Derive the capacitance of the cable of length  $l$ . What happens if the dielectric is removed, if the charge  $Q$  remains constant?

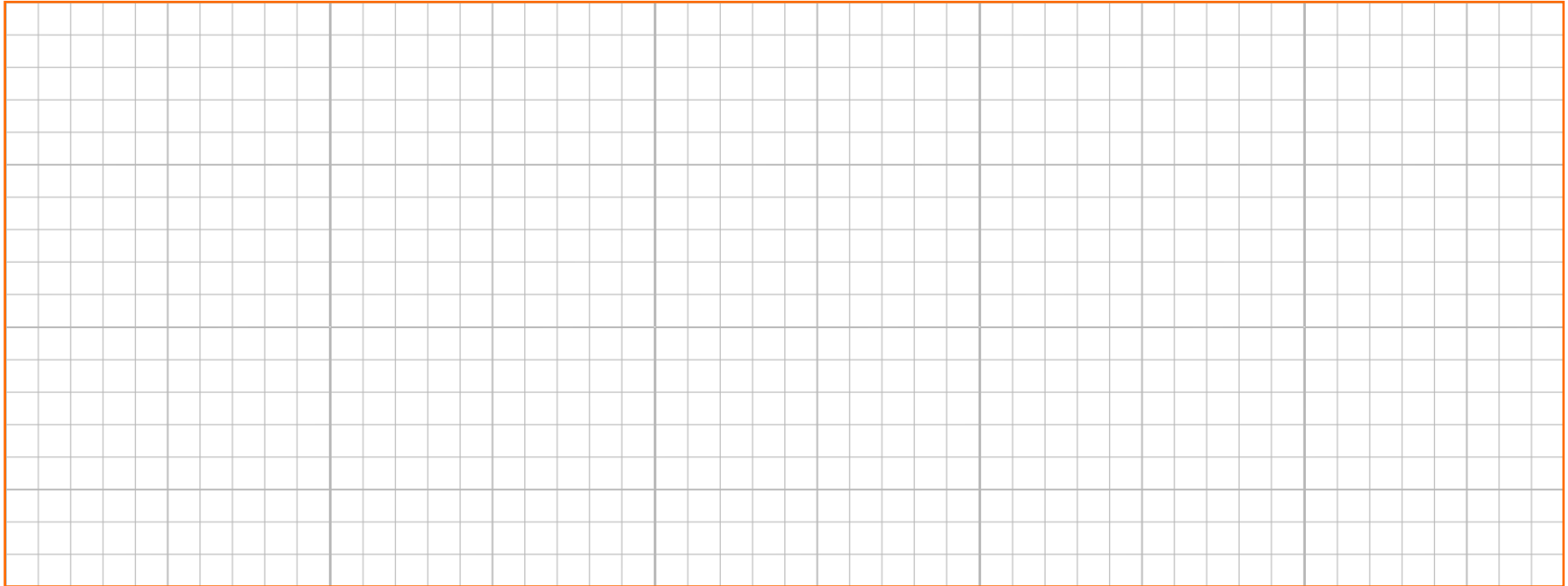


## 2.5 Field applications: energy storage and conductivity

Homework: Cylindrical capacitor / Coaxial cable with non-constant permittivity



Now analyse the same setup but with the dielectric being a function  $\varepsilon_r = r \varepsilon_1$ . What do you notice?



## 2.5 Field applications: energy storage and conductivity thws

### Dielectric strength

- The **dielectric strength**  $\|\vec{E}_{\max}\|$  describes the maximum field that can be applied before spark discharge.

$$\|\vec{E}\| \leq \|\vec{E}_{\max}\|$$

- The value depends on the geometry, the dielectric between the electrodes, the maximum voltage  $U_{\max}$  and the operating temperature.

- ┐ For a parallel plate capacitor with homogeneous field

$$U_{\max} = \|\vec{E}_{\max}\| d$$

- ┐ For any arbitrary shape with inhomogeneous field

$$U_{\max} = \frac{Q_{\max}}{C} = \frac{\varepsilon \|\vec{E}_{\max}\| A}{C}$$

- *Note: Typically, the rated voltage of a capacitor is 20 % lower than the maximum voltage  $U_{\max}$ .*

| Dielectric strength $\ \vec{E}_{\max}\ $ | $\frac{\text{MV}}{\text{m}}$ |
|------------------------------------------|------------------------------|
| Vacuum                                   | 10                           |
| Air                                      | 2 – 3                        |
| Porcelain                                | 30 – 35                      |
| Plastics                                 | 25 – 150                     |
| SiO <sub>2</sub> (on a chip)             | 600                          |

Table with the dielectric strength of some selected materials.

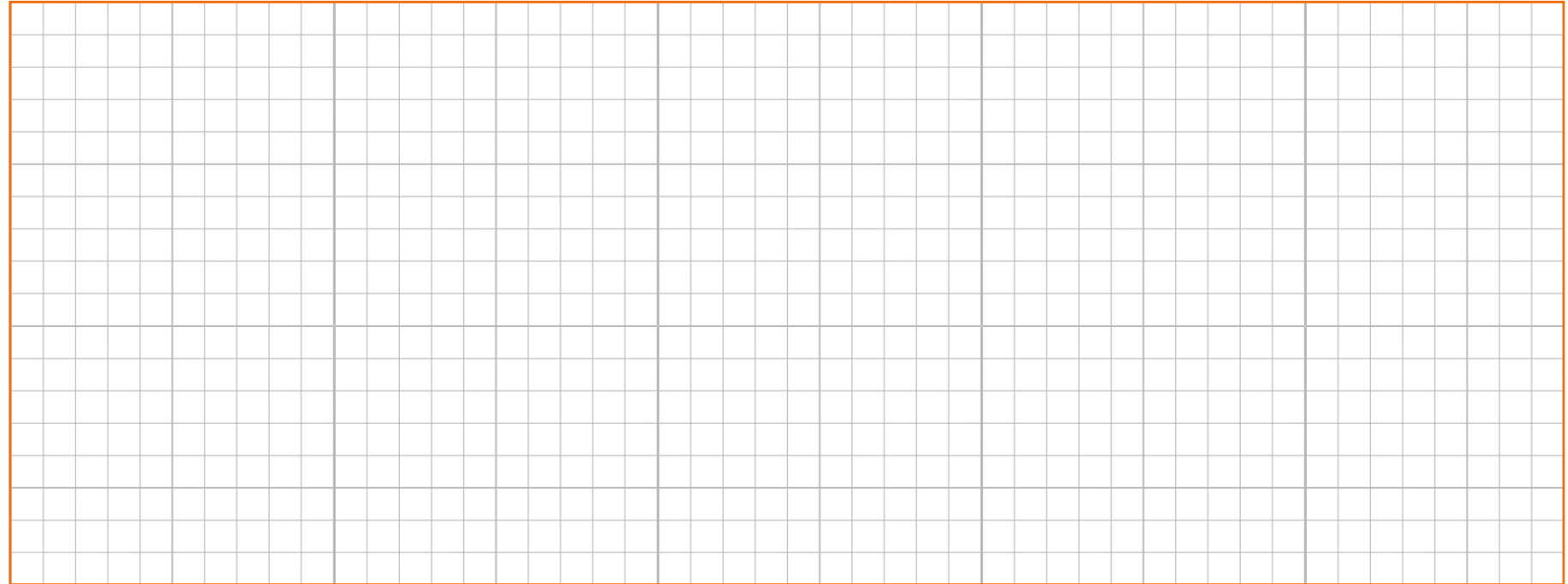
## 2.5 Field applications: energy storage and conductivity

### Exercise: Dielectric strength

|      |          |
|------|----------|
| EXAM | Typical  |
| • —  | Exam     |
| • —  |          |
| • —  | Question |



Calculate the maximum voltage  $U_{\max}$  of a parallel plate capacitor (plate distance  $d = 23 \mu\text{m}$ ) and a cylindrical capacitor (inner radius  $r_1 = 1 \text{ mm}$ , outer radius  $r_2 = 2.72 \text{ mm}$ ) with a PP-dielectric ( $\|\vec{E}_{\max}\| = 100 \text{ MV/m}$ ).



## 2.5 Field applications: energy storage and conductivity thws

### Mobile charge

- An **electric field** acts on a **charge** according to

$$\vec{F} = Q \vec{E}$$

- ┐ If the charge  $Q$  is stationary, “nothing” happens
- ┐ If the **charge is mobile**, it will **accelerate**

$$\vec{F} = Q \vec{E} = m \vec{a}$$

- In regular media (metals, conductors) charges reach a **steady state**, the **drift velocity**

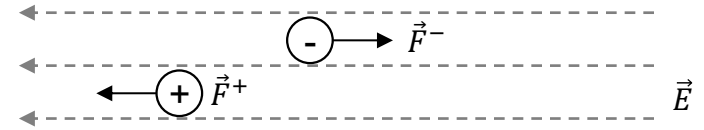
$$\vec{v}_d = \mu \vec{E},$$

where the mobility  $\mu$  factors in how quickly the charge can move through a medium.

- The mobility is limited by defects and lattice interaction
  - ┐ **Defect**: vacancy, interstitial, smaller/larger atom
  - ┐ Lattice vibration or **phonon**



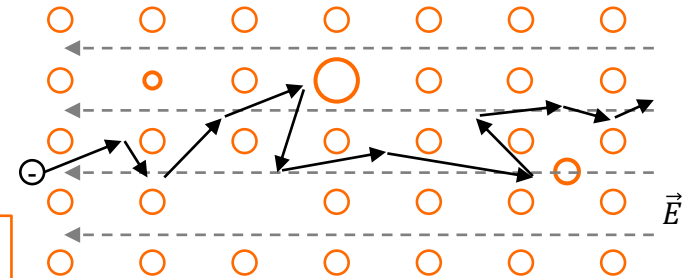
*Because of phonons the conductivity of metals decreases with temperature!*



*Model of the electric current in a metallic wire.*



*A positive charge moves in field direction and an equally negative charge opposite to field. Both yield the same net-current.*



*Simple point defects in a metal.*

# 2.5 Field applications: energy storage and conductivity

## Current density

 For metals typically  
 $n \approx 10^{23} \text{ cm}^{-3}$  and  
 $\mu \approx 10^{-3} \frac{\text{m}}{\text{Vs}}$

- A number  $n$  of charges  $Q$  constitute **current density**  $\vec{j}$ .

$$\vec{j} = n |Q| \vec{v}_d = n \mu |Q| \vec{E} = \kappa \vec{E} \quad [j] = \text{A/m}^2,$$

with **the conventional drift velocity**  $\vec{v}_d$  and **conductivity**  $\kappa = n \mu |Q|$ ,  $[\kappa] = 1/\Omega \text{ m}$ .

- The **electric current**  $I$  is **current density** through an area:

$$I = \vec{j} \cdot \vec{A}$$

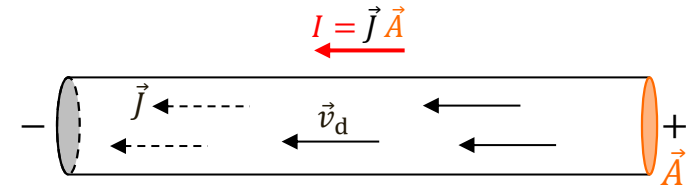
- In most elements (wire, resistor)  $\vec{j}$  and  $\vec{A}$  are **aligned**, hence

$$I = \pm \|\vec{j}\| \|\vec{A}\|$$

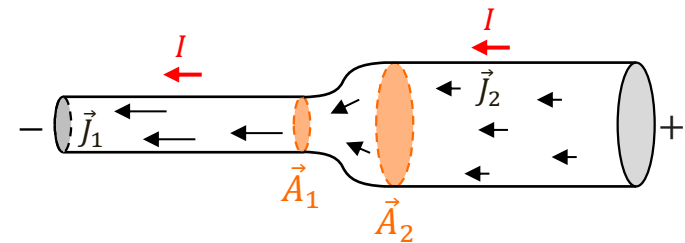
- The law of continuity (a. k. a. **KCL**) requires that **current**  $I$  **through elements** in series is **constant** independently of shape, size and other properties of each element.
- As the current is constant, the ratio of the **current densities** for elements in series is **inverse** to ratio of their **areas**.

$$\frac{\|\vec{j}_1\|}{\|\vec{j}_2\|} = \frac{\|\vec{A}_2\|}{\|\vec{A}_1\|}$$

 **Conventional velocity:**  
*electrons actually move  
in the opposite direction!*

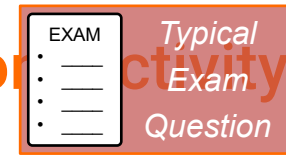


*Model of the electric current in a metallic wire.  
The current density is in direction of  
conventional current!*



*Model of the electric current in a metallic wire  
of varying diameter.*

## 2.5 Field applications: energy storage and conductivity



### Exercise: Copper wire in a field 1/2

Consider a copper wire ( $n = 8.47 \cdot 10^{22} \text{ cm}^{-3}$ ,  $\mu = 44 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$  at room temperature) in an electric field of magnitude  $\|\vec{E}\| = 5 \text{ mV m}^{-1}$ . Calculate the conductivity  $\kappa$ , drift velocity  $\|\vec{v}_d\|$  and magnitude of current density  $\|\vec{j}\|$ . What is the current in a wire of area  $A = 1 \text{ mm}^2$



## 2.5 Field applications: energy storage and conductivity thws

### Ohm's law revisited

- In an **element** of area  $A$  and length  $l$  with **homogeneous field**  $\vec{E} = E = \text{const.}$  and **current density**  $\vec{j}$  **parallel to the wire**, the current density can be expressed as electric current

$$I = \kappa \frac{A}{l} U,$$

Because  $I = \pm \|\vec{j}\| \|\vec{A}\|$   
and  $\|\vec{j}\| = \kappa \|\vec{E}\| = \kappa \frac{U}{l}$

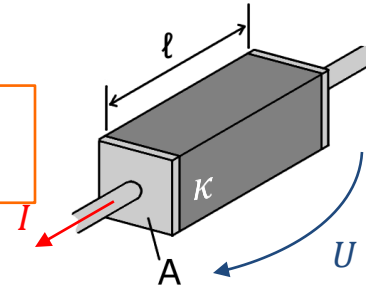
- By substituting  $R = \frac{1}{\kappa} \frac{l}{A}$ , **Ohm's law** can be derived.

$$R = \frac{U}{I}$$

- Sometimes the **inverse of the conductivity**, the **resistivity**  $\rho = 1/\kappa$  is utilized. The resistance is then defined as

$$R = \rho \frac{l}{A}$$

- *Note: We'll analyse the conductivity of different types of conductors in chapter 4.*

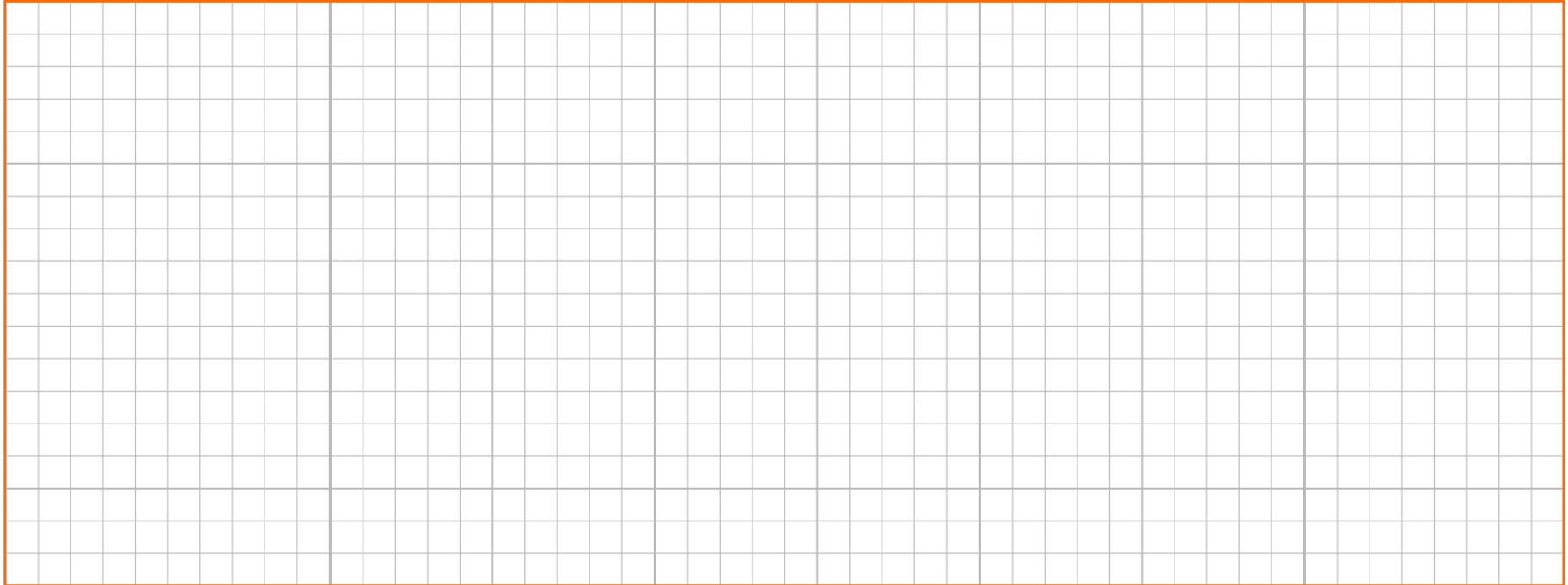


*A piece of resistive material with electrical contacts.*

## 2.5 Field applications: energy storage and conductivity thws

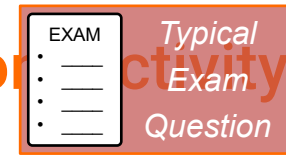
### Homework: Current density to current conversion

Derive the equation  $I = \kappa \frac{A}{l} U$  from the definition of the charge density  $\vec{J} = \kappa \vec{E}$  in a wire. *Note: Use  $J = \|\vec{J}\|$  and  $A = \|\vec{A}\|$  for better readability.*

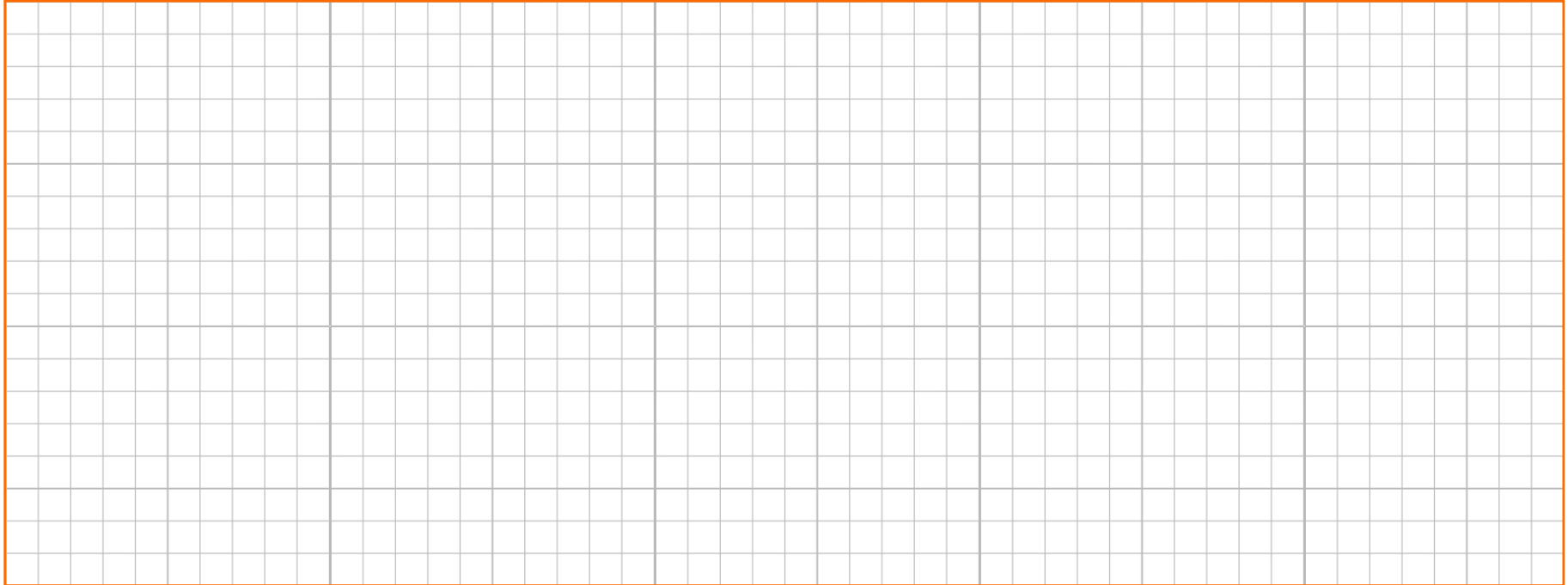


## 2.5 Field applications: energy storage and conductivity

### Exercise: Copper wire in a field 2/2



Consider the copper wire from before. Please determine the current if the wire has a diameter  $d = 2$  mm. Also calculate the resistance  $R$  of a wire element of length  $l = 20$  m and the voltage across this element.



## Summary

### — Capacitance & Energy

└ Capacitance from geometry:  $\vec{E}$  via Gauss' law  $\rightarrow U = \int_{r_1}^{r_2} \|\vec{E}\| \, dr \rightarrow C = \frac{Q}{U} = \frac{\varepsilon}{\int_{r_1}^{r_2} \frac{1}{A} dr}$

└ Energy:  $W = \frac{1}{2} C U^2 = \frac{1}{2} \frac{Q^2}{C} = \frac{1}{2} Q U; w_V = \frac{1}{2} \varepsilon \|\vec{E}\|^2$

└ Dielectric strength:  $\|\vec{E}\| \leq \|\vec{E}_{\max}\|; U_{\max} = \|\vec{E}_{\max}\| d$

### — Current density & Conductivity

└ Microscopic:  $\vec{v}_d = \mu \vec{E}$  (drift velocity),  $\vec{J} = n |Q| \vec{v}_d = \kappa \vec{E}$  ( $\kappa = n Q \mu$ ),  $I = \pm \|\vec{J}\| \|\vec{A}\|$  (in a wire)

└ Ohm's Law: From  $\vec{J} = \kappa \vec{E} \rightarrow I = \kappa U \frac{A}{l} \rightarrow R \frac{U}{I} = \rho \frac{l}{A}$ , where  $\rho = \frac{1}{\kappa}$

└ Properties: Mobility  $\mu$  limited by defects/phonons;  $\kappa$  decreases with  $T$  in metals

└ **Key Insight:** Ohm's law emerges from microscopic charge motion, linking field theory to circuits

## 2.5 Field applications: energy storage and conductivity

## 4.0.0

- Initial version SoSe26

## 4.0.1

- Minor typographic changes, error correction

## 4.0.2

- Symbol for the electric potential changed from  $\phi$  to  $\varphi$  in one exercise

## 4.1.0

- Slide on conductors moved from chapter Gauss to chapter Materials

## 4.2.0

- Mostly visual corrections

## 4.3.0

- Example on dielectric strength updated with non-homogeneous field and visual correction in complete version

## 4.4.0

- Error correction

## 4.5.0

- Hint for exam relevant questions added